

Genetic and physical interactions reveal overlapping and distinct contributions to meiotic double-strand break formation in *C. elegans*

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Marilina Raices, Fabiola Balmir, Nicola Silva, Wei Li, McKenzie K Grundy, Dane K Hoffman, Elisabeth Altendorfer, Carlos Jaime Camacho, Kara A Bernstein, Monica P Colaiácovo, Judith L Yanowitz 

Magee-Womens Research Institute, Pittsburgh, United States • Department of Biology, Masaryk University, Brno, Czech Republic • Tsinghua University Medical School, Beijing, China • Department of Microbiology and Molecular Genetics, University of Pittsburgh School of Medicine, UPMC Hillman Cancer Center, Pittsburgh, United States • Department of Genetics, Blavatnik Institute, Harvard Medical School, Boston, United States • Department of Computational and Systems Biology, University of Pittsburgh School of Medicine, Pittsburgh, United States • Department of Biochemistry and Biophysics, University of Pennsylvania, Penn Center for Genome Integrity, Philadelphia, United States • Department of Obstetrics, Gynecology and Reproductive Sciences, University of Pittsburgh School of Medicine, Pittsburgh, United States

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This study combines genetic, cell biological, and interaction data to propose a model of meiotic double-strand break regulation in *C. elegans*. **Solid** evidence supports the main conclusions, while by nature of a screening-type study, more may be needed to solidify speculations in future studies. Yet, comprehensive cataloging of the physical and genetic interactions of factors required for meiotic double-strand break is **useful** information for the field.

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Abstract

Double-strand breaks (DSBs) are the most deleterious lesions experienced by our genome. Yet, DSBs are intentionally induced during gamete formation to promote the exchange of genetic material between homologous chromosomes. While the conserved topoisomerase-like enzyme Spo11 catalyzes DSBs, additional regulatory proteins—referred to as “Spo11 accessory factors”—regulate the number, timing, and placement of DSBs during meiotic prophase ensuring that SPO11 does not wreak havoc on the genome. Despite the importance of the accessory factors, they are poorly conserved at the sequence level suggesting that these factors may adopt unique functions in different species. In this work, we present a detailed analysis of the genetic and physical interactions between the DSB factors in the nematode *Caenorhabditis elegans* providing new insights into conserved and novel functions of these proteins. This work shows that HIM-5 is the determinant of X-chromosome-specific crossovers and that its retention in the nucleus is dependent on DSB-1, the sole accessory

factor that interacts with SPO-11. We further provide evidence that HIM-5 mediates interactions with the different accessory factors sub-groups, providing insights into how components on the DNA loops may interact with the chromosome axis.

Introduction

One of the seminal events during meiosis is the formation of crossovers (CO) during the first meiotic division. While COs create new combinations of alleles through the exchange of genetic material between homologs, they also establish physical connections that ensure their proper alignment on the meiotic spindle and their subsequent segregation to opposite ends of the spindle. In the absence of COs, homologs segregate randomly, leading to gametes with altered chromosome numbers and subsequent fetal aneuploidies. In humans, aneuploidy is observed in 1 of 160 live births (Dungan, 2002 [↗](#)) and in >50% of miscarriages (Hassold et al., 2007 [↗](#)) underscoring the importance of proper CO formation.

A necessary early step in CO formation is the creation of DNA double-strand breaks (DSBs) catalyzed by the widely conserved Spo11 enzyme (Bergerat et al., 1997 [↗](#); Keeney et al., 1997 [↗](#)). Although Spo11 is essential for DSB formation, it does not function alone: additional proteins regulate the timing, placement, and number of DSBs (Cole et al., 2010 [↗](#); Hinman et al., 2021 [↗](#); Keeney, 2008 [↗](#); Vrielynck et al., 2021 [↗](#)). In *Saccharomyces cerevisiae*, where DSB formation has been best characterized, at least nine other proteins interact with Spo11 to regulate its recruitment and activation (Lam & Keeney, 2015 [↗](#); Martini et al., 2006 [↗](#)). *In vivo* and *in vitro* physical interaction studies have revealed that these proteins form several subcomplexes. Briefly, the RMM complex (Rec114, Mer2, and Mei4) is loaded on chromosomes early in prophase and is required for Spo11 binding to sites of DNA cleavage (J. Li et al., 2006 [↗](#); Sasanuma et al., 2007 [↗](#)). Activation of Mer2 requires phosphorylation by cyclin-dependent kinase (CDK) (Henderson et al., 2006 [↗](#)), suggesting that Mer2 links DSB formation to the progression of the meiotic program (Arora et al., 2004 [↗](#); Henderson et al., 2006 [↗](#); J. Li et al., 2006 [↗](#); Maleki et al., 2007 [↗](#)). Rec102 and Rec104 form another subcomplex that facilitates Spo11 dimerization and its association with DSB sites (Arora et al., 2004 [↗](#); Kee & Keeney, 2002 [↗](#); Prieler et al., 2005 [↗](#); Sasanuma et al., 2007 [↗](#)). Ski8 binds Spo11 and is important for Spo11 nuclear localization (Arora et al., 2004 [↗](#)). Finally, the Mre11–Rad50–Xrs2 (MRX) complex is also required for DSB formation but requires all the other DSB proteins to associate with DSB sites (Bouaert et al., 2021 [↗](#)). Despite their importance, most of these proteins are poorly conserved at the amino acid level although homologs have been identified for several of these in organisms ranging from plants to worms to mice (Cole et al., 2010 [↗](#); Kumar et al., 2010b). Importantly, a recent study of plants has presented a similar model for DSB induction as in yeast, yet the specific players and details differ (Vrielynck et al., 2021 [↗](#)) highlighting the value of comparative studies to understand the evolutionary history of the DSB machinery in metazoans.

In vitro biochemical studies have shown that *C. elegans* SPO-11 is monomeric but that even when it is able to bind dsDNA, it does not exhibit DNA cleavage activity (Yeh et al., 2017 [↗](#)). This observation strongly supports the hypothesis that cofactors are needed for DSB induction. DSB-1 and DSB-2, are homologs of the conserved Rec114 protein (Tesse et al., 2017), and DSB-3, a homolog of Mei4 (Hinman et al., 2021 [↗](#)). These factors localize to chromosomes during early meiotic prophase and promote the activity of the SPO-11 complex (Rosu et al., 2013 [↗](#); Stamper et al., 2013 [↗](#); Hinman et al., 2021 [↗](#)), functioning analogously to their counterparts in other species (Kumar et al., 2010a [↗](#), 2015 [↗](#), 2018 [↗](#); Li et al., 2006 [↗](#); Maleki et al., 2007 [↗](#)). *him-17*, *mre-11*, and *rad-50* are also essential for the introduction of breaks in this organism (Chin & Villeneuve, 2001 [↗](#); Hayashi et al., 2007 [↗](#); Hinman et al., 2021 [↗](#); Reddy & Villeneuve, 2004 [↗](#); Stamper et al., 2013 [↗](#)).

Other genes, including *xnd-1*, *him-5*, *rec-1*, *cep-1*, and *parg-1* have more limited roles in break induction. *him-5* is fascinating in that it is required to ensure the complement of DSBs on the X chromosome (Janisiw et al., 2020 [↗](#); Meneely et al., 2012 [↗](#); Rosu et al., 2013 [↗](#)). In the region of the germ line where DSBs are made, the X chromosome is transcriptionally quiescent and packaged in a heterochromatin-like state (Fong et al., 2002 [↗](#); Kelly et al., 2002 [↗](#)). By contrast, the autosomes are transcriptionally active and replete with histone modifications associated with open chromatin. It is thought that the heterochromatic-like state of the X presents a barrier to DSB formation necessitating the evolution of proteins to overcome this obstacle (Wagner et al., 2010 [↗](#)). *xnd-1* transcriptionally regulates *him-5* to promote breaks on the X (McClendon et al., 2016 [↗](#)), but it also restricts histone acetylation, a function that ensures the proper timing of DSB induction (Wagner et al., 2010 [↗](#)). In the absence of *xnd-1*, there is precocious formation of DSBs as monitored by the accumulation of the HR strand-exchange protein RAD-51 in transition zone nuclei (Gao et al., 2015 [↗](#); McClendon et al., 2016 [↗](#); Wagner et al., 2010 [↗](#)).

CO positioning is altered in all situations where DSB levels are affected, but the *rec-1* mutation has a profound effect on the distribution of meiotic COs (Rose & Baillie, 1979 [↗](#)) with only a minor reduction in total DSB numbers (Chung et al., 2015 [↗](#)). Prior analysis of *rec-1* and *him-5* found these genes to be paralogs that function cooperatively to ensure wild-type DSB levels (Chung et al., 2015 [↗](#)). Interestingly, while loss of *rec-1* preferentially affects CO distribution, loss of *him-5* reduces DSB numbers nearly in half, severely delays DSB induction (until mid-late pachytene), and changes CO distribution (Meneely et al., 2012 [↗](#)). *him-5* has redundant functions with *cep-1* for both break formation and inhibition of NHEJ (Mateo et al., 2016 [↗](#)), revealing roles for *him-5* in coordinating DSB induction with downstream repair.

In this study, we expand the analysis of the genetic interactions between the SPO-11 accessory factors in *C. elegans* and provide the first protein-protein interaction studies of these factors using co-immunoprecipitation and yeast-two hybrid (Y2H). We use these interaction data to propose how the DSB formation is initiated in *C. elegans* and compare and contrast this to models proposed in yeast and plants (Bouaert et al., 2021 [↗](#), Vrielynck et al., 2021 [↗](#)). Our results allow us to begin to elucidate the regulatory events that control the localization, timing, and placement of meiotic DSBs in *C. elegans*.

Results

HIM-5 is the essential factor for meiotic breaks in the X-chromosome

Three of the DSB proteins appear to have a biased effect on X chromosome CO formation: null alleles of *xnd-1* and *him-5* predominantly show a loss of CO intermediates on the X, increases in univalent X chromosomes at diakinesis in half or almost all nuclei, respectively, and an increase in XO male offspring resulting from nondisjunction of the X chromosome during the meiotic division; the hypomorphic *him-17(e2806)* mutation shows increased incidence of one unattached homolog pair in diakinesis and an increase in male production consistent with an X-specific defect (Reddy & Villeneuve, 2004 [↗](#)). To determine if these genes function through a common underlying mechanism, we first performed yeast two-hybrid (Y2H) assessing interactions with both high and low stringency. We observed a strong Y2H interaction between XND-1 and HIM-17, and a weak one between HIM-17 and HIM-5 (Figure 1A [↗](#) and Figure S1 [↗](#)). To validate these interactions, we also performed immunoprecipitations (IPs) of HIM-17::GFP followed by mass spectrometry (MS, Figure S1B [↗](#)). This identified XND-1, as well as 16 additional strong interactors of HIM-17::GFP (Table S2 [↗](#)). Moreover, endogenous XND-1 was identified in western blots of IPs performed independently using 3xHA-tagged HIM-17 epitopes, indicating a robust interaction between these

proteins (**Figure 1B** and S1C). In contrast, HIM-5 was not detected in either the HIM-17::GFP co-IPs or IP/MS, suggesting that the interaction observed by Y2H either exists very transiently or was bridged by a protein expressed in yeast.

Prior studies from our group indicate that *xnd-1* regulates HIM-5 transcriptionally since ectopic expression of *him-5::GFP* using the *pie-1* germline promoter, but not expression using the endogenous *him-5* promoter, rescues the X chromosome nondisjunction defects conferred by the *xnd-1* mutation (Wagner et al., 2010; Meneely et al., 2012; McClendon et al., 2016). Since RNA-Seq studies showed that *him-17* regulates many germline genes including *him-5* (Carelli et al., 2022), we set out to determine if ectopic expression of *him-5* could also rescue *him-17* null mutants. Similar to what we observed with *xnd-1* mutants, *Ppie-1::him-5* — but not *Phim-5::him-5* — rescued the X chromosome defects seen by the increase of XO male offspring from XX hermaphrodites (known as the high incidence of males or “Him” phenotype) (35% of males in *him-17(ok424)* vs. 7.02% in *Ppie-1::him-5;him-17(ok424)*) (**Figure 1C**). Consistent with the lack of rescue, there was no statistically significant change in univalent formation in *Phim-5::him-5;him-17(ok424)* (**Figure S2**). These results cannot be explained by HIM-5 protein levels which are lower in *Ppie-1::him-5* than in *Phim-5::him-5* (McClendon et al., 2016). The lack of rescue from *Phim-5::him-5* supports the conclusion that HIM-17 directly regulates *him-5* expression. Since the *pie-1* promoter appears to be independent of HIM-17, the *Ppie-1::him-5* construct can—at least partially—bypass this regulation and suppress the *him-17* mutant phenotype. These findings suggest that both XND-1 and HIM-17 both promote X chromosome DSB and CO formation by regulating *him-5* expression. Moreover, *xnd-1* and *him-17* have distinct roles in DSB/CO formation since only null alleles of *him-17*, but not *xnd-1* or *him-5*, confer a severe loss of DSBs (Meneely et al., 2012; Reddy & Villeneuve, 2004; Wagner et al., 2010), while only *xnd-1* mutations influence the timing of DSB induction/repair (Wagner et al., 2010; McClendon et al., 2016).

We note that ectopic HIM-5 expression strongly rescued the embryonic lethality conferred by the *him-17* null allele (**Figure 1C**). While a small fraction of the lethality is attributed to X chromosome nondisjunction, most of the lethality is a consequence of autosomal missegregation. Thus, even though *him-5* null mutant animals exhibit a profound loss of DSBs/COs on the X chromosomes with rare CO defects on autosomes (Meneely et al., 2012), *him-5* expression can restore CO formation to both autosomes and X chromosomes. This strongly supports a general role for HIM-5 in CO formation and further indicates that HIM-5 is sufficient to restore COs on the X chromosome.

To further characterize the interaction between XND-1 and HIM-17, we also sought to determine whether they colocalize on meiotic chromosomes using confocal microscopy. Prior studies have shown that HIM-17 associates with meiotic chromatin (Reddy & Villeneuve, 2004). Co-staining the HIM-17::HA transgenic line with anti-HA and anti-XND-1 antibodies showed that HIM-17, like XND-1, is enriched on autosomes (**Figure 1D**; Wagner et al., 2010). Both proteins showed no overlap with the chromosome axes (marked with HTP-3 (**Figure 1D**) or DAPI (**Figure S3B**), although we cannot rule out that a small fraction of the population, either below the level of detection or transiently, localizes to the axes. These immunolocalization studies also show a strong juxtaposition of HIM-17 and XND-1, with no detectable overlap between their staining patterns in germ lines from the mitotic region through late pachytene, when XND-1 disappears from the nucleus (**Figure 1E** and **Figure S3B**). Structural modeling of the XND-1 and HIM-17 interaction (Abramson et al., 2024; Comeau et al., 2004) provides a potential explanation for this observation: HIM-17 is a very large protein, and the HA-tag positioned at its amino terminus is predicted to be located at least 180 Å away from the protein:protein interaction interface (**Figure 1F**).

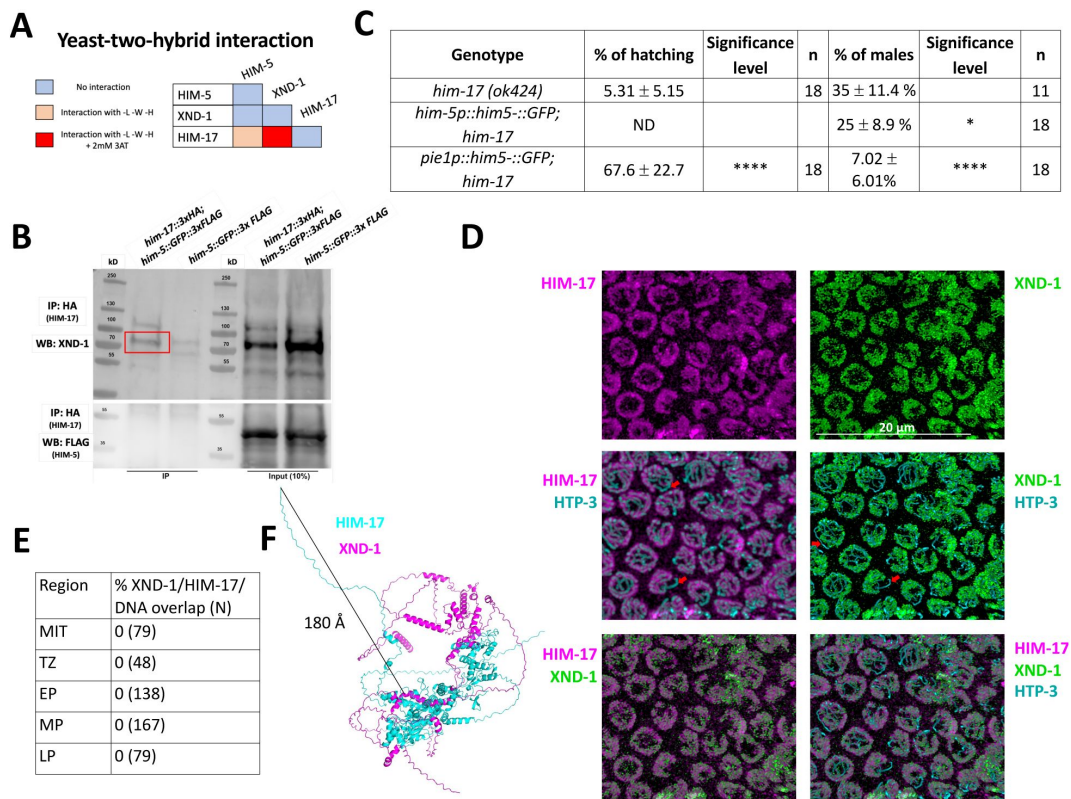


Figure 1

XND-1 and HIM-17 interact and are predominantly localized on DNA loops.

A) Summary of the yeast two-hybrid assay results. The complete set of results is given in [Figure S1A](#). **B)** Western blot analysis of endogenous XND-1 (top) and HIM-5::GFP::FLAG (bottom) on HA pull-downs performed in *him-17::3xHA; him-5::GFP::3xFLAG* strain. Red box indicates the XND-1 band. Analysis was performed in biological duplicates. Specificity of the anti-XND-1 antibody is shown in [Figure S3A](#). **C)** Quantification of embryonic viability (hatching rates) and frequencies of male offspring among the progeny of the indicated genotypes. Data are shown as mean ± SD; * $p < 0.05$, **** $p < 0.0001$, ND= not determined, n= number of P_0 parents whose brood was examined. **D)** Representative confocal images of mid-pachytene stage nuclei showing that juxtaposed XND-1 and HIM-17 localization away from the chromosome axes (stained with anti-HTP-3). The X chromosome is not stained by either XND-1 and HIM-17 (Red arrows). Scale bars = 20µm. **E)** Quantification of the pairwise overlap between XND-1, HIM-17, and the chromosome axes. **F)** AlphaFold3 model of XND-1-HIM-17 dimer. The black line shows the 180 Å distance between the HA tag in HIM-17 and the XND-1 protein which is recognized by the polyclonal antibody.

DSB-1 mediates interactions with SPO-11 and promotes HIM-5 nuclear localization

Since HIM-5 appears to be central to DSB/CO formation on the X chromosome and the presence of DSB-1 and DSB-2 are required for all meiotic breaks, we next sought to determine whether HIM-5 interacts with DSB-1, its paralog DSB-2, or DSB-3. We found that DSB-1 and DSB-2 show robust Y2H interactions and that DSB-1 and DSB-3 exhibit weak Y2H interactions (**Figure 2A** [↗](#), **Figure S1A** [↗](#)), confirming results from a prior study (Hinman et al., 2021 [↗](#)). We also found that HIM-5 shows weak Y2H interactions with DSB-1 (**Figure 2A** [↗](#)). We confirmed the interaction between DSB-1 and HIM-5 with co-IP, pulling down GFP::DSB-1 and probing for HIM-5::3xHA by Western blotting (**Figure 2B** [↗](#)).

We next wanted to understand the functional relevance of the interaction between DSB-1 and HIM-5. To this end, we examined the localization of HIM-5 proteins in the *dsb-1* mutant background. Co-staining of the chromosome axis (anti-HTP-3) and either HIM-5::GFP transgene (anti-GFP) or endogenously-tagged HIM-5::3xHA (anti-HA) revealed that HIM-5 nuclear localization begins just prior to the transition zone in both wild type and *dsb-1* mutants (**Figure 2C** [↗](#) and **Figure S4A** [↗](#), arrows). However, whereas in wild type, HIM-5 protein remains nuclear throughout early-mid pachytene and become further enriched on meiotic chromatin; in *dsb-1* mutants, HIM-5 nuclear localization is lost (**Figures 2C** [↗](#) and **Figure S4B** [↗](#)). From this change in HIM-5 protein distribution, we infer that DSB-1 (and/or its DSB-promoting function) is required to retain HIM-5 in meiotic nuclei.

Since HIM-5 levels could also be affected by *dsb-1* and/or *dsb-2*, we next asked whether ectopic expression of HIM-5 could suppress defects in bivalent formation due to the loss of *dsb-1* or *dsb-2*. As shown in **Figure 2D** [↗](#), ectopic HIM-5 reduced the number of univalents in both mutant backgrounds. Since a small fraction of *dsb-2*; *Ppie-1::him-5* animals exhibited 6 or 7 DAPI bodies, we assayed whether the transgene could improve hatching rates and reduce male production in *dsb-2* (**Figure 2E** [↗](#)). Whereas the *Ppie-1::him-5* transgene has no impact of hatching and male frequencies, the more highly expressed *Phim-5::him-5* transgene (McClendon et al., 2016 [↗](#)) conferred a mild suppression of the embryonic lethality. This ability to suppress hatching-but not male production-further supports a role for HIM-5 in regulating total DSB numbers.

Genetic interactions define four functional groups that contribute to DSB/CO formation

Having established interactions between subsets of the SPO-11 accessory factors in *C. elegans*, we next wanted to perform more extensive genetic analyses between the known DSB factors. To this end, we created double mutant combinations of alleles with partial DSB defects. These included either weak loss-of-function alleles of essential DSB genes (*him-17*, *mre-11*) or null alleles of genes that partially reduce DSBs (*cep-1*, *dsb-2*, *rec-1*, *him-5*, and *parg-1*). Diakinesis oocytes were analyzed in the single and double mutants by whole-mount fixation and DAPI staining followed by confocal microscopy and 3D visualization. The chromosomes are highly condensed at diakinesis, and six bivalents can be detected in wild-type worms. Univalents, fusions, and DNA fragments can be seen in mutants with defects in DSB formation or their repair into a CO, respectively (Chin & Villeneuve, 2001 [↗](#); Guo et al., 2022 [↗](#); Rosu et al., 2013 [↗](#)). We hypothesized that double mutant strains carrying mutations in genes that function in different aspects of DSB induction would show more severe defects and have an increase in univalent formation compared to single mutants. On the other hand, genes that function together or at the same step of break induction should exhibit no or only mild enhancement of the DSB defects.

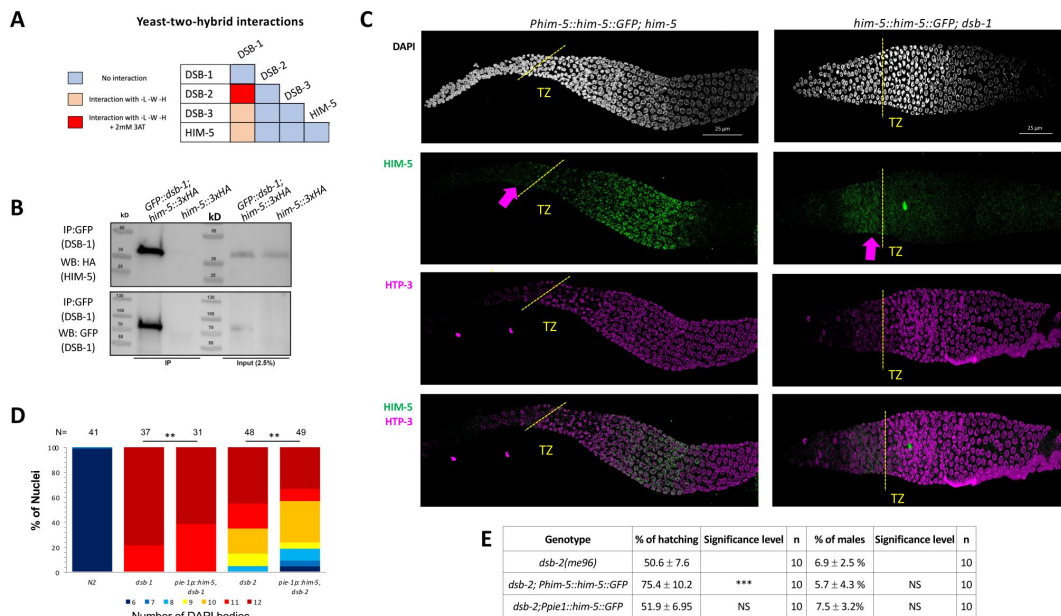


Figure 2

DSB-1 associates with HIM-5 and regulates its localization

A) Summary of yeast two-hybrid results with *dsb-1*, *dsb-2*, *dsb-3* and *him-5*. The complete set of results is given in **Figure S1A**. **B**) Western blot analysis of anti-GFP pull-downs performed in *GFP::dsb-1; him-5::3XHA* strain showing co-IP of HA-tagged HIM-5 proteins. Analysis was performed in biological duplicates. **C**) Immunofluorescence analysis of HIM-5 localization in *Phim-5::him-5::GFP; him-5* and *Phim-5::him-5::GFP; dsb-1* backgrounds. DNA (DAPI, gray), HIM-5::GFP (anti-GFP, green), and chromosome axes (anti-pHTP-3, magenta). pHTP-3 staining delineates entrance to the transition zone (marked dotted yellow lines), while arrows indicate the pre-meiotic localization of HIM-5. **D**) Quantification of DAPI-stained bodies at diakinesis for the indicated genotypes. Colors correspond to the number of DAPI-stained bodies shown in the key below for *N2*, *dsb-1*, and *dsb-2* mutants expressing *Ppie-1::him-5::GFP*. Sample sizes (N) are indicated. Statistical significance for comparisons between groups is shown at the top (**p < 0.01). **E**) Quantification of embryonic viability (hatching rates) and frequencies of male offspring among the progeny of the indicated genotypes. Data are shown as mean $\pm$ SD; NS stands for not significant; *** p < 0.001, n = number of P₀ parents whose brood was examined.

Double mutant analysis was precluded with *dsb-1* and *dsb-3* since the single mutants already show a complete absence of breaks (Hinman et al., 2021 [DOI](#); Stamper et al., 2013 [DOI](#)). By contrast, *dsb-2* mutant nuclei manifest an age-dependent loss in DSB capacity (Rosu et al., 2013 [DOI](#); Stamper et al., 2013 [DOI](#)) allowing us to analyze interactions between *dsb-2* and *cep-1*, *parg-1*, *him-17*, and *rec-1* (Figure 3A [DOI](#) and Figure S5A-D [DOI](#)). As 1-day-old adults, *dsb-2* single mutants had ~50% of diakinesis nuclei that contained 6 bivalents, ~30% with 7 DAPI-stained bodies (5 bivalents and 2 univalents), and ~20% with 8 or more DAPI-stained bodies. Single mutants of *cep-1*, *parg-1*, and *rec-1* each had >90% of diakinesis nuclei with 6 bivalents, consistent with prior studies (Chung et al., 2015 [DOI](#); Janisiw et al., 2020 [DOI](#); Mateo et al., 2016 [DOI](#)). In animals carrying *him-17(e2806)* mutation, a weak allele of the otherwise essential locus for DSB formation, only ~50% of mutant nuclei had 6 bivalents; the other 50% had 7 DAPI-stained bodies, reflecting the requirement for this gene in CO formation on the X chromosome (Reddy & Villeneuve, 2004 [DOI](#)). When *dsb-2* was combined with each of these mutations, an increased number of DAPI-stained bodies and a concomitant decrease in the number of chiasmata were observed (Figure 3A [DOI](#), Figure S5 [DOI](#)). These results strongly suggest that *dsb-2* functions cooperatively with all four loci for the regulation of meiotic CO formation.

Given that *rec-1*, *parg-1*, and *cep-1* fall in a separate functional group from *dsb-2*, we next addressed whether these three genes function together or independently. All double mutant permutations of *rec-1*, *parg-1*, and *cep-1* were analyzed, and in each case, the single and double mutants were nearly indistinguishable from the single mutants, i.e., 6 bivalents (Figure S5L-M [DOI](#)). Therefore, we infer that these genes likely function to control the same step in DSB/CO induction (Figure 3B [DOI](#)). Consistent with this interpretation, when *rec-1*, *parg-1*, or *cep-1* were combined with other DSB gene mutations, including *him-17*, they had similar effects, and the numbers of univalents were increased (Figures S54A, C, D for Group 1; Figure S5E-G [DOI](#) for Group 3; and Figure S5J [DOI](#), K for Group 4). Similar results have previously been reported from our laboratories for double mutants with *him-5* (Chung et al., 2015 [DOI](#); Janisiw et al., 2020 [DOI](#); Mateo et al., 2016 [DOI](#)).

mre-11 functions in both DSB formation and subsequent DNA end resection (Chin & Villeneuve, 2001 [DOI](#)), the latter as part of the MRE11-RAD50-NBS1/Xrs2 (MRN/X) resection complex, the other proteins of which are encoded by the worm *rad-50* and *nbs-1* genes (Girard et al., 2018 [DOI](#); Hayashi et al., 2007 [DOI](#)). Null mutations in *mre-11* and *rad-50* completely abrogate DSB induction. However, we were able to take advantage of a separation-of-function allele, *mre-11(iow1)*, that cannot perform its end-resection functions but can induce DSBs (Yin & Smolikove, 2013 [DOI](#)). The *iow1* mutation causes chromosome fusions to appear in diakinesis oocytes resulting from the aggregation of unprocessed DNA ends. While this allele makes enough DSBs to cause aggregation of all chromosomes, we hypothesized that it might have a minor impairment in DSB induction that would be revealed when combined with other DSB mutations. If this were the case, the double mutants would make fewer breaks, leading to fewer chromosome fusions and an increase in the number of DAPI-stained bodies/univalents. Consistent with this hypothesis, *mre-11(iow1)* double mutants with each of the other DSB mutations led to fewer fusions and more DAPI-stained bodies (Figure 3A [DOI](#), Figure S5H-K [DOI](#)). Thus, we propose that *mre-11*, and by extension *rad-50*, function in their own branch of the DSB/CO pathway.

Surprisingly, a unique phenotype was observed in a subset of the *cep-1*; *mre-11* double mutants. Our prior studies showed that *cep-1* functions redundantly with *him-5* to both ensure DSB formation and promote downstream homologous recombination (HR) repair by preventing non-homologous end joining (NHEJ) (Mateo et al., 2016 [DOI](#)). In that study, we reported that *cep-1(lg12501)* is a separation-of-function allele that is defective in DSB formation. In the *cep-1(lg12501)*; *mre-11(iow-1)* double mutants, we observed mixed phenotypes: some nuclei had 12 DAPI-stained bodies, while others exhibited an exacerbation of the fusion defect (Figure S6 [DOI](#)). We infer that the univalents indicate a synergistic effect between *mre-11* and *cep-1* for DSB formation; the increase in fusions, on the other hand, may suggest that *cep-1(lg12501)* is not fully wild-type for the DNA repair functions of CEP-1.

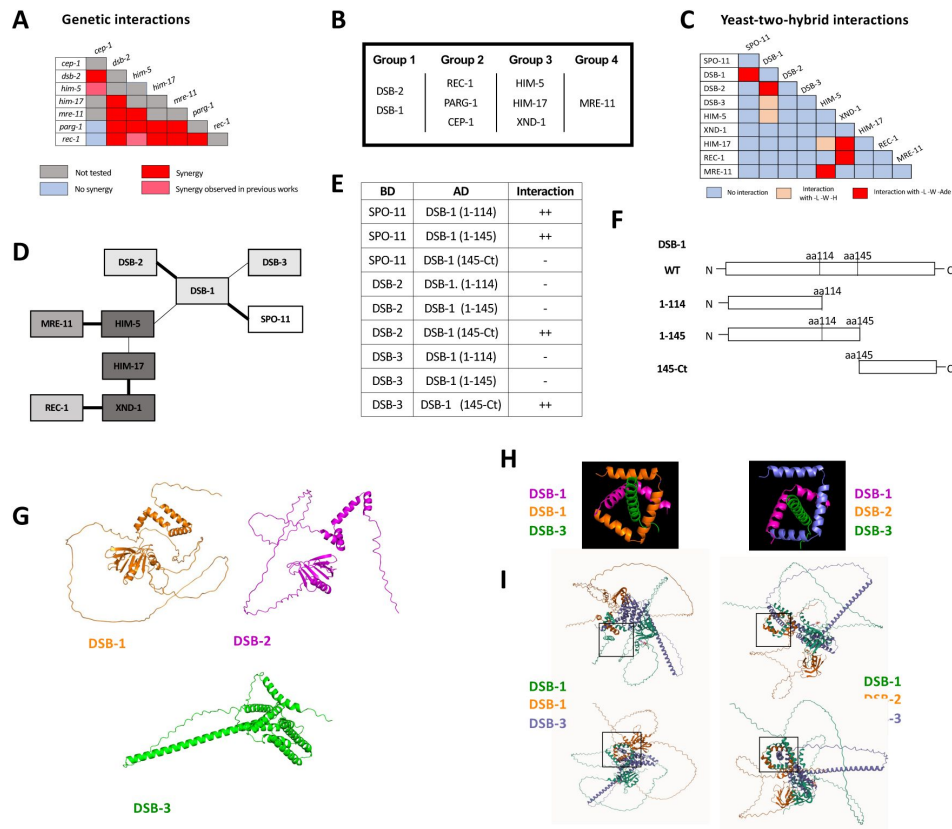


Figure 3

Genetic and protein-protein interaction studies show evidence of SPO-11 accessory protein sub-complexes in *C. elegans*.

A) Summary of the genetic interaction results. The complete set of results is given in [Figure S5](#). **B)** Genetic groupings based on data presented here and previously published ([Chung et al., 2015](#); [Mateo et al., 2016](#); [McClendon et al., 2016](#); [Janisiw et al., 2020](#)). **C)** Summary of yeast two-hybrid results. The complete set of results are shown in [Figure S1A](#). **D)** Schematic representation of the interaction network based on Y2H interactions. Thick line: Strong interaction; thin line: weak interaction. **E, F)** Yeast-two-hybrid interactions of SPO-11, DSB-2, and DSB-3 with different DSB-1 sub-domains (schematic in **F**). Details on DSB-1 sequence, protein structure, and deletions used in these experiments, are found in [Figure S8](#). **G)** AlphaFold2 predictions of DSB-1, DSB-2 and DSB-3 protein structures. **H, I)** Predicted interaction domains of DSB-1, DSB-2 with DSB-3 based on homology with the yeast proteins. The DSB-1/-1/-3 trimer is thermodynamically more stable than the DSB-1/-2/-3 trimer, consistent with the subordinate role for DSB-2 in break induction in young animals.

While each of the genes that we analyzed has reported roles in DSB formation, we wanted to confirm that the observed increases in univalents in the double mutants were in fact due to impairment in DSB induction and not to deficits in later steps of CO formation. Therefore, we relied on an established assay to distinguish between these possibilities: we asked whether the addition of exogenous breaks from gamma irradiation (IR) could suppress the univalent phenotype by serving as a surrogate for SPO-11-induced DSBs (Dernburg et al., 1998 [↗](#)). Recent work has shown that 10Gy IR is able to produce up to ~20 DSBs per meiotic nucleus (Lascarez-Lagunas et al., 2023 [↗](#)), which is sufficient to ensure a CO on each chromosome (Mets & Meyer, 2009 [↗](#)). We analyzed the number of DAPI bodies with and without IR in a subset of the double mutants where we observed synergistic effects (*dsb-2*; *parg-1*, *dsb-2*; *rec-1*, and *rec-1*; *him-17*). When each of these double mutant strains was exposed to IR, the wild-type phenotype of 6 bivalents was largely restored (Figure S7 [↗](#)), supporting the conclusion that the CO deficiency observed is triggered by defects in DSB formation.

Protein-protein interactions between the DSB regulatory factors

The genetic analyses described above defined four functional groups that contribute to break formation (Figure 3B [↗](#)). Using Y2H, we next set out to determine whether interactions between and within groups could explain how the different complexes come together to regulate DSB induction (Figure 3C-D [↗](#), Figure S1A [↗](#)). CEP-1 showed auto-activation in both vectors and therefore was removed from further analysis. DSB-1, HIM-17, and REC-1 also showed self-activation in the binding domain vector, therefore, only the activation domain fusions were analyzed. Pairwise interactions between all remaining combinations were then analyzed. HIM-5 displayed the most promiscuous Y2H interactions, with strong interactions seen with MRE-11 and weak interactions as already discussed with HIM-17 and DSB-1. XND-1 apart from interacting with HIM-17, also showed strong Y2H interactions with REC-1. A recent study provided Y2H evidence supporting the idea that SPO-11 interacts with DSB-1 (Hinman et al., 2021 [↗](#)). Critically, our work supports and extends this observation to show that DSB-1 is the only known accessory factor that interacts with SPO-11 in Y2H assays, cementing its role as a key determinant for DSB induction and explaining its role in feedback controls (Stamper et al., 2013 [↗](#); Hinman et al., 2021 [↗](#); Guo et al., 2022 [↗](#)).

To analyze more in-depth the Y2H interactions between DSB-1 and DSB-2, DSB-3, and SPO-11, we looked for protein domains in DSB-1 that could be involved in these interactions. Interestingly, DSB-1 has 5 “sticky” helices (helices 2-6 in Figure S8 [↗](#)) that have the potential to be implicated in protein-protein interactions. We first deleted all of these helices together by introducing a deletion after L114 (DSB-1(1-114) in Figures 3E [↗](#), F). We found that this truncated version of DSB-1 loses the ability to interact with DSB-2 and DSB-3 but is still able to associate with SPO-11. Moreover, when we only deleted the helices 3-6 but kept the helix 2 (DSB-1(1-145)), we obtained the same results, suggesting that the Y2H interactions between DSB-1 and DSB-2/DSB-3 is mediated by the C-terminal helices 3-6 and not helix 2. To prove this, we created a truncated DSB-1 protein that only contains amino acids from P145 to the stop codon (DSB-1(145-Ct)). Consistent with the deletion analysis, we observed that the C-terminal domain is sufficient to promote associations with DSB-2 and DSB-3. This domain was also not sufficient to promote the DSB-1/SPO-11 interaction (Figure 3E [↗](#)). These experiments complement the results published by Villeneuve lab where they observed that the DSB-1 N-terminus is the domain necessary for the DSB-1/SPO-11 (but not for the DSB-1/2/3) interaction (Hinman et al., 2021 [↗](#)).

We next used AlphaFold and structural modeling to investigate whether the *C. elegans* REC114/MEI4 homolog could assemble in the same fashion as their distant orthologues (Claeys Bouuaert et al., 2021 [↗](#); Daccache et al., 2023 [↗](#)) in which two helices of REC114 dimer surround a MEI4 helix. DSB-1 is predicted to form a stable homodimer that can assemble into a highly stable trimeric complex with DSB-3 threaded through the middle (Figure 3G-I [↗](#)) analogous to what is predicted for Rec114-Mei4 (Daccache et al., 2023 [↗](#)). By contrast, DSB-2 cannot form homodimers, although it can replace one of the DSB-1 monomers to form the DSB-1:DSB-2:DSB-3 ternary

complex. Free energy calculations (Champ and Camacho, 2007) show that the DSB-1:DSB-2:DSB-3 complex is less stable (−3.8 kcal/mol) than a DSB-1:DSB-1:DSB-3 complex consistent with the more limited role for DSB-2 in break formation, at least in young worms (Rosu et al., 2013; Stamper et al., 2013).

Discussion

Across species, accessory factors collaborate with SPO-11 to regulate various aspects of DSB formation, including the timing, placement, number, and persistence of DSB induction, as well as the chromatin architecture at and around the break site (Arora et al., 2004; Cole et al., 2010; Hinman et al., 2021; Kumar & de Massy, 2010; Panizza et al., 2011; Sasanuma et al., 2007; Sommermeyer et al., 2013; Vrielynck et al., 2021). Here we present genetic, biochemical, and cytological studies to develop a working interaction model for DSB regulation in *C. elegans*. The *C. elegans* DSB complex appears to be comprised of four sub-groups (Figure 3B and Figure 4) interconnected by their association with HIM-5. Our results suggest two important roles for HIM-5: first as the linchpin between the whole DSB complex, thereby ensuring the coordination of the timing, placement, and number of DSBs (described in more detail below); and second, as the determinant in X chromosome COs.

DSB-1 and DSB-2 are both orthologs of the budding yeast Rec114 protein but DSB-1 has a more central role in DSB induction (Rosu et al., 2013; Stamper et al., 2013) which we posit can be explained by the increased stability of the DSB-1:DSB-1:DSB-3 trimer compared to a DSB-1:DSB-2:DSB-3 trimer. The direct association between SPO-11 and DSB-1 but none of the other worm accessory factors points to DSB-1/REC114 as the hub for the integration of break inducing signals in this organism. In yeast and *Arabidopsis*, accessory proteins are associated indirectly with Spo11/SPO11 through interactions with the TopVI/MTOPVIB subunit (Arora et al., 2004; Kee & Keeney, 2002; Maleki et al., 2007; Robert et al., 2016; Vrielynck et al., 2021). These differences between species underscore how conserved DSB proteins have been co-opted to regulate different aspects of break induction.

While DSB-1/-2/-3 are critical for CO formation, they predominantly localize on DNA loops with little to no overlap (Hinman et al., 2021). Instead, a small fraction of these three proteins appear to associate with the DNA axis, a localization that is consistent with the “tethered loop model” for break induction in yeast and plants (Kee et al., 2004; Maleki et al., 2007; Panizza et al., 2011; Sommermeyer et al., 2013; Tsai et al., 2020). Two accessory factors, MRE-11 and PARG-1 are known interactors of the axis protein HTP-3 (Janisiw et al., 2020). Both proteins interact either directly or indirectly with HIM-5, which physically associates with DSB-1, thereby providing a mechanism to bring the loop-associated proteins to the axis (Figure 4).

MRE-11 and PARG-1 also have roles in both break formation and DNA repair. For PARG-1, we show here that it functions together with REC-1 and CEP-1 based on the following data: synthetic CO defects were seen in *cep-1;him-5*, *rec-1;him-5*, and *parg-1;him-5* (Chung et al., 2015; Mateo et al., 2016 and this paper), but not in *rec-1;cep-1*, *rec-1;parg-1*, and *cep-1;parg-1* double mutants; *dsb-2* mutant phenotypes are exacerbated by the loss of *rec-1*, *cep-1* (Figure S5), and *parg-1* (Janisiw et al., 2020). Interestingly, both CEP-1 and PARG-1 have additional roles in promoting homologous recombination-mediated repair and/or preventing alternative repair pathways (Janisiw et al., 2020; Mateo et al., 2016). We envision that CEP-1, REC-1, and PARG-1 function together to promote DSBs and their repair into COs at the chromosome axis. Since we also showed that the DSB defects in *mre-11* mutant animals were enhanced by loss of *parg-1* or *rec-1* (Figure S5), we favor a model in which MRE-11 and PARG-1 independently interact with the chromosome axis to ensure robust DSB formation and/or repair. It is possible that MRE-11, which is essential for COs, facilitates the tethering of chromosome loops to the axis, but further structural studies of these complexes are required.

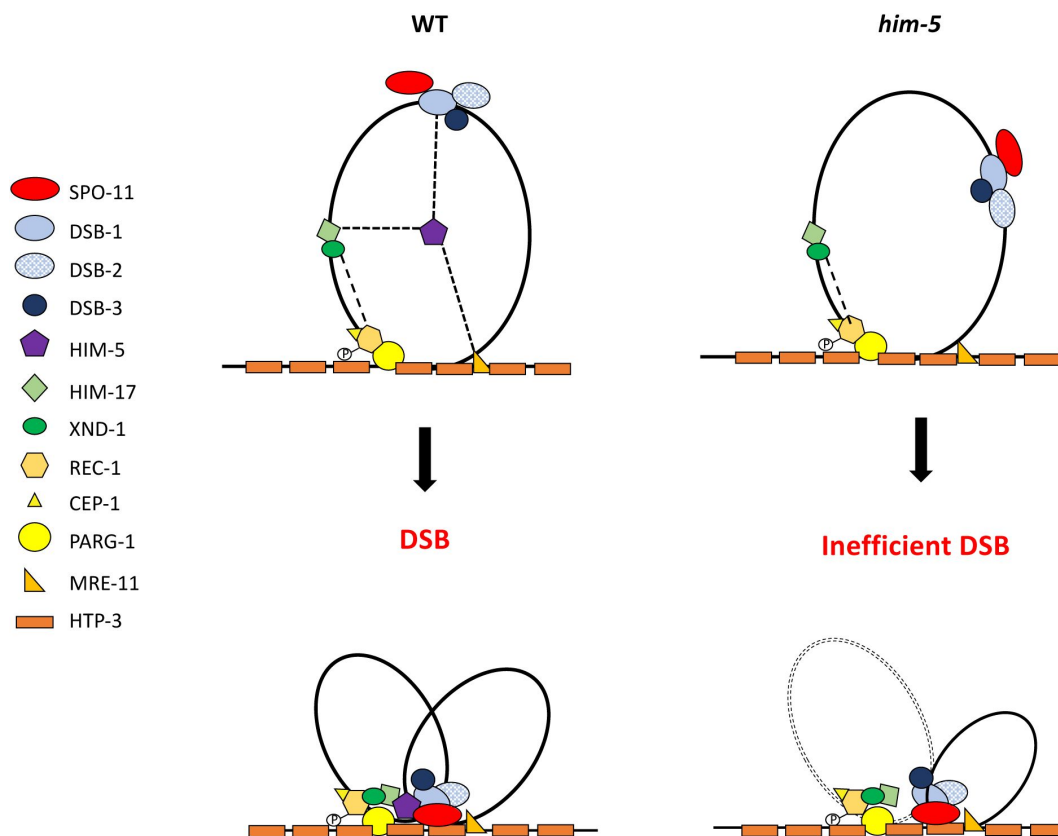


Figure 4

Speculative model of the DSB formation complex in *C. elegans*

Synthesizing prior work and data herein, we propose a model to explain the known interactions and localization patterns of the DSB regulatory factors in *C. elegans*. DSB-1/-2/-3 and HIM-17-XND-1 subcomplexes are initially found on DNA loops. The phosphorylation of REC-1 by CDK activates DSB formation, allowing HIM-5 through its association with MRE-11 and PARG-1 (which both associate with HTP-3) to efficiently bring the DSB-1/-2/-3 complex with SPO-11 to the chromosome axis. This recruitment to the axes by HIM-5, facilitates the coupling of DSBs to downstream repair by the MRN complex and others. In the absence of HIM-5, DSBs/COs on autosomes do not occur efficiently and crossover sites are shifted to the gene-rich third of each chromosome, presumably because DSB-1/-2/-3/SPO-11 cannot be as efficiently recruited to the proper sites partially defined by HIM-17 and XND-1. We assume that recruitment to the chromosome axes in the absence of HIM-5 is likely stochastic or mediated by as yet unknown protein interactions. On the X chromosome, the absence of HIM-5 prevents the formation of most DSBs.

The defects observed in *cep-1;him-5* (Mateo et al., 2016 [↗](#)), and *rec-1;him-5* (Chung et al., 2015 [↗](#)) are both exacerbated by maternal age, similar to *dsb-2*. We envision that inputs from both HIM-5 and DSB-2 converge on DSB-1 to ensure robust DSB formation across the lifespan. This would also explain the synthetic defects in DSB formation between *dsb-2* and each of the components in the *him-5* interaction network. DSB-1 is required for chromatin association of DSB-2 but not vice versa, although DSB-1 levels are reduced in older *dsb-2* mutant animals (Rosu et al., 2013 [↗](#); Stamper et al., 2013 [↗](#)). We show here that DSB-1 is also required for nuclear retention of HIM-5. While the functional significance of this interaction awaits isolation of separation-of-function mutations that abrogate their interaction, it is tempting to speculate the signals that initiate meiotic induction regulate this interaction.

TIMING of BREAKS

Studies from multiple systems have elucidated mechanisms to coordinate the timing of DSBs with the completion of meiosis, e.g. by CDK and Ddf4 phosphorylation in yeast (Murakami & Keeney, 2008 [↗](#); Carelli et al., 2022). In worms, REC-1 was previously shown to be a target of CDK and therefore it was hypothesized that such a phosphorylation event would be critical to ensure that break induction occurs after the completion of S phase (Chung et al., 2015 [↗](#)). REC-1 and HIM-5 evolved from a gene duplication event in the *C. elegans* lineage; whereas the ancestral gene in related nematodes has features of both proteins-although it shares greater homology with HIM-5 (Chung et al., 2015 [↗](#)). Thus, we hypothesize that the single ancestral REC-1/HIM-5 protein would be subject to CDK phosphorylation, which would then promote the assembly of the complexes at the chromosome axis to promote break formation.

Timing also appears to be critical for CO outcomes. Studies in yeast show different outcomes for early and late breaks (Joshi et al., 2015 [↗](#)) and recent studies in worms indicate that later breaks are needed for CO formation (Hicks et al., 2022). We previously showed that *xnd-1* mutants have altered DSB kinetics, showing elevated levels of DSB formation in leptotene/zygotene (McClendon et al., 2016 [↗](#); Wagner et al., 2010 [↗](#)). This function of XND-1 is mediated, at least in part, by its effect on chromatin architecture. One possibility is that the physical interaction that we show here between XND-1 and REC-1 couples the completion of S phase to the activation of breaks within the correct chromatin environment, thus ensuring their timely formation. In the absence of XND-1, increases in global histone acetylation (Gao et al., 2015 [↗](#); Wagner et al., 2010 [↗](#)) might allow SPO-11 easier access to the DNA. Another possibility is that upon phosphorylation of REC-1 by CDK-1, REC-1 through PARG-1 transiently brings XND-1 (the timer) to the axis, an event that restrains DSB formation.

Chromatin and Meiotic Gene Transcription

We provide strong evidence for direct interactions between XND-1 and HIM-17, both by Y2H and co-IP. HIM-17 appears to have a role in transcriptional regulation of DSB factors. Here we show that ectopic expression of HIM-5 from the *pie-1* promoter, but not overexpression from the endogenous *him-5* promoter, suppressed *him-17* phenotypes. This is similar to the rescue we saw of *xnd-1* (McClendon et al., 2016 [↗](#)) and suggests that both proteins contribute to the expression of *him-5*. This supports findings that *him-17* regulates the expression of ~300 germline-enriched genes, including *him-5*, *rec-1*, and *dsb-2*, (Carelli et al., 2022) whose concomitant loss would be expected to abrogate break induction (Chung et al., 2015 [↗](#) and **Figure 3A** [↗](#) and **Figure S5** [↗](#)). While its only role may be in transcriptional regulation, a growing literature and data presented herein suggest a more direct role for HIM-17 in meiosis. We presented Y2H data for a weak interaction between HIM-17 and HIM-5. Although these interactions were not confirmed by IP, it remains possible that these complexes are transient associations, are unstable during IP, and/or are not abundant. Together with observed changes in chromatin structure and both DSB and CO distribution in *him-17* mutants (Nadarajan et al., 2021 [↗](#); Reddy & Villeneuve, 2004 [↗](#)), it is tempting to speculate that together with XND-1, HIM-17 helps to create an environment

permissive for SPO-11 binding and/or break formation. XND-1 and HIM-17 may function together in this role or redundantly, perhaps through XND-1 interacting with REC-1 and HIM-17 interacting with HIM-5, or in related species through interactions with the single REC-1/HIM-5 paralog.

X/Autosome differences in DSB formation

One of the conundrums arising from our data is in the function of the HIM-5 protein. *him-5* was identified based on its preferential impact on CO formation on the X chromosome (Broverman & Meneely, 1994 [\[1\]](#); Hodgkin et al., 1979 [\[2\]](#)). The X-specific phenotypes conferred by *him-5* mutation could suggest that the X chromosome is simply more sensitive to a reduction in the number of active SPO-11 complexes; the fact that mutations in *dsb-2* also reduce DSBs but without an X chromosome bias argues against this interpretation. We show herein that HIM-5 is sufficient to ensure that COs occur on the X chromosome. Ectopic expression of *him-5* can substantially rescue both the autosomal and X chromosomal defects of *him-17* null mutations – despite expression levels lower than endogenous *him-5* (McClendon et al., 2016 [\[3\]](#)). The X chromosome-specific defects of *him-5* mutations might arise due to defects in recruitment of DSB-1-2-3/SPO-11 to specific sites on the X or to an impaired ability to coordinate the assembly of the DSB machinery on the—late-replicating (Jaramillo-Lambert et al., 2007 [\[4\]](#))—axis of the X chromosome, although other models can be envisioned.

Our data and a growing literature insinuate a more general and central role for HIM-5 in DSB formation and downstream CO repair: 1) *rec-1;him-5* double mutants give an age-dependent severe loss of DSBs (like *dsb-2* mutants) suggesting that the ancestral function of the protein may have a more profound effect on break formation (Chung et al., 2015 [\[5\]](#)); 2) total DSBs are reduced to nearly 50% of wild type and are severely delayed in formation in *him-5* mutants (Meneely et al., 2012 [\[6\]](#)); 3) a recent study reports role for *him-5* in CO formation in males where the X chromosome does not engage in CO exchange (Engbrecht et al., 2025 [\[7\]](#)); 4) HIM-5 is required for the normal distribution of COs on the autosomes (Meneely et al., 2012 [\[6\]](#)); and 5) redundantly with CEP-1, HIM-5 promotes DSB formation and HR-mediated repair and inhibits NHEJ (Mateo et al., 2016 [\[8\]](#)). Our data herein show that HIM-5 interacts not only with DSB-1, but also components of the other functional groups that regulate DSB induction. Together, we believe this data supports a role for HIM-5 in bridging the accessory factors with SPO-11 to ensure the proper timing, placement and number of breaks.

Concluding Remarks

Our Y2H and IPs results show very strong protein-protein interactions between the different DSB factors, however, their localization along the gonads does not frequently coincide suggesting temporal and spatial regulation throughout meiosis. Further studies are necessary to elucidate how these interactions arise and change *in vivo* and how these are integrated with both cell cycle and regulatory feedback controls.

Methods

Culture and strains

Worms were cultured on MyoB plates (Burns et al., 2006 [\[9\]](#)) seeded with OP50 at 20°C, unless otherwise noted (Brenner 1974). Mutant strains used in this study are listed in Supplemental Material, Table S1 [\[10\]](#). All strains were derived from the wild-type Bristol strain N2.

Chromosome morphology analysis of diakinesis oocytes

The numbers of DAPI-stained bodies present in diakinesis oocytes were assessed in intact adult hermaphrodites at day 1 of adulthood (24 hours post-L4, unless otherwise specified). Adults were fixed in Carnoy's fixative solution (60% ethanol, 30% chloroform, and 10% glacial acetic acid) and stained with 4',6-diamidino-2-phenylindole (DAPI) (10 mg/ml stock diluted 1:50,000 in 1X Phosphate-buffered saline; PBS) for 15 minutes in a humid chamber. Worms were mounted in Prolong Gold with DAPI, cured overnight at room temperature, and stored at 4 °C prior to imaging either on a Nikon A1r confocal microscope or Leica Stellaris 5 confocal with an integrated White Light Laser (WLL). Diakinesis images were procured as 0.2µm per plain Z-stacks and visualized using Volocity 3-D Imaging Software for Nikon images and LAS X for Leica images. Fisher tests were performed to analyze the DAPI-stained chromosome morphology data in diakinesis oocytes. This approach was used to compare the distribution of DAPI-stained bodies, focusing on differences in the frequency of univalents and bivalents. Statistical analyses were conducted using R software, with a significance threshold set at $p < 0.05$.

γ-irradiation

Worms of specified genotypes were exposed to 10Gy of γ-irradiation using a ^{137}Cs source (Gammacell1000 Elite; Nordion International). Analysis of diakinesis oocytes by DAPI staining was performed as described above at 28 h post-irradiation.

Immunostaining

Gonads from day 1 adults of the appropriate genotypes were dissected in 1X sperm salts (50 mM PIPES pH 7.0, 25 mM KCl, 1 mM MgSO_4 , 45 mM NaCl, and 2 mM CaCl_2) with 1 mM levamisole and fixed in 1% paraformaldehyde diluted in 1X sperm salts for 5 min in a humid chamber. Slides were then frozen on a metal block on dry ice for at least 10 min prior to flicking off the cover slip and immersing in 100% ethanol at -20° C for 2 min. Slides were then washed 3X for 10 min each in PBSTB [1XPBS with 0.1% Tween and 0.1% bovine serum albumin (BSA)], and prior to overnight incubation with primary antibody also diluted in PBSTB. Primary antibodies were as follows: mouse anti-GFP 1:500 (Invitrogen Cat. #33-25—Lot 683990A), rabbit anti-phosphoHTP-3^{S285} 1:2000 (Das et al., 2022 [DOI](#)), mouse anti-HA 1:1000 (mouse anti-HA-tag (6E2) Cell Signaling Cat. #23675, Lot 5); guinea pig anti-XND-1, 1:10,000 (Wagner et al., 2010 [DOI](#)). Overnight incubations were performed at 4° C except for anti-HA which was performed at room temperature. The next day, slides were washed 3X in PBSTB for 10 min each and incubated with Alexa-conjugated secondary antibodies from Molecular Probes: anti-mouse Alexa 488, anti-rabbit Alexa 633, anti-guinea pig Alexa 568 (all diluted 1:2000 in PBST/ BSA 0.1%). Incubation was for 2 h at room temperature in the dark. Slides were then washed 2 X 10 min in PBSTB, and 1 X 10 min with DAPI (10 mg/ml stock diluted 1:50,000 in 1X PBS). Slides were mounted in Prolong Diamond with DAPI and put in the dark to cure overnight before imaging. Images were acquired on a Leica Stellaris 5 confocal and Lightning image processing during acquisition. Z-stacks were acquired with 0.2 micron or smaller step size and visualized in 3D with Leica LAS X software.

Yeast two-hybrid assay

cDNAs were amplified from germline-specific cDNA libraries and were cloned by Gibson Assembly into the GAL4 activating domain (pGAD-C1) and binding domain (pGBD-C1)– expressing vectors. Plasmids with the desired cDNA were verified by Sanger sequencing prior to use. Plasmids were co-transformed into the *PJ69-4a* yeast strain and selected by growth on SC-LEU-TRP medium. Three to five transformants were grown to early log phase ($\text{OD}_{600} = 0.2$), and 5 µL of culture was spotted onto selection plates:

- SC-LEU-TRP (loading control, selecting for plasmid presence)
- SC-LEU-TRP-HIS (reporter *HIS3* expression, indicating interaction)

- SC-LEU-TRP-ADE (more stringent selection for strong interactions)

Plates were incubated at 30°C for 72 hours and then imaged. The experiment was performed in triplicate, and empty vectors in the activating and binding domains were included as negative controls. Data are presented as spots on a plate and are from three to four individual transformants per interaction tested and are not individual colonies. The experiment was repeated in triplicate from different transformations.

Co-Immunoprecipitation and Western Blot

In order to perform co-immunoprecipitation experiments, synchronized *GFP::dsb-1; him-5::3xHA* and *him-5::3xHA* or *him-17::3xHA;Phim-5::him-5::GFP::3xFLAG* and *Phim-1::him-5::GFP::3xFLAG* strains were grown until young adult stage (24h post-L4) and nuclear protein fractionation was executed as previously shown (Silva et al.; 2014). The chromatin-bound fraction was generated upon Benzonase digestion (25U/100 µl of extract for 1h at 4°C). 2 mg of nuclear extract (nuclear-soluble and chromatin-bound fractions were pooled together) was incubated with Agarose GFP traps (Chromotek) or anti-HA Affinity Matrix (Roche) in Buffer D (20mM HEPES pH 7.9, 150mM KCl, 20% glycerol, 0.2mM EDTA, 0.2% Triton X-100 and 1x complete Roche inhibitor) overnight at 4°C. The following day, the agarose beads were recovered by centrifugation at 7500 rpm for 2' at 4°C and extensively washed with Buffer D at room temperature. After the final wash, the beads were resuspended in 40 µl of 2x Laemmli Buffer and boiled for 10', after which they were spun at maximum speed for 1' and the whole eluate was loaded on a precast 4-20% gradient acrylamide gel. Proteins were transferred on a nitrocellulose membrane for 90 minutes at 4°C at 100V and blocked for 1h at room temperature in 1x TBST containing 5% BSA. Mouse monoclonal anti-HA (Cell Signaling), chicken polyclonal anti-GFP (AbCam), mouse monoclonal anti-FLAG HRP-conjugated (Sigma), and polyclonal anti-XND-1 (Wagner et al., 2010) antibodies were diluted in blocking solution at 1:1000, 1:5000, 1:2000 and 1:2500 respectively and left to incubate overnight at 4°C. Washes were performed in 1x TBST and anti-mouse, anti-chicken, and anti-guinea pig HRP-conjugated secondary antibodies (ThermoFisher) were diluted 1:10,000 in 1x TBST containing 5% milk for 1h at room temperature. After several washes in 1xTBST, the membrane was incubated with Clarity Max ECL (BioRad) and imaged with a G-Box (Syngene).

Immunoprecipitation and Mass spectrometry analysis of HIM-17::GFP

IPs were performed on stock AV280: *unc-119(e2498) III; him-17(ok424) V; meIs5[him-17::GFP +unc-119(+)]*. Preparation and freezing of synchronized worms were carried out as in Cheeseman et al., (2004) except that worms were grown on 8P plates seeded with the *E. coli* strain NA22 at room temperature. Worm lysis was carried out by grinding worm pellets in liquid nitrogen and subsequent sonication (Diagenode) was performed with the following settings: 3x 5 cycles on high with 30 seconds on and 30 seconds off. Lysate was centrifuged for 45 minutes at 15000 rpm at 4°C. To avoid unspecific protein binding the worm lysate was pre-cleared by incubation to uncoupled agarose beads (Chromotek, bab-20) for 1 hour at 4°C. Pre-cleared worm lysate was incubated with GFP-trap (Chromotek, gta-20) for 3 hours at 4°C. Beads were washed 3 times with lysis buffer containing proteinase inhibitors (Roche) and 7 times with lysis buffer without proteinase inhibitors. Proteins were eluted by adding 40 µl 2xLSB and boiling for 5 min at 95°C. Protein precipitation was carried out using ProteoExtract Protein Precipitation Kit (EMD Millipore, 539180-1KIT) according to the manufacturer's protocol and submitted for MS analysis to the Taplin Biological Mass Spectrometry Facility. The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository (Perez-Riverol et al., 2022) with the dataset identifier PXD057629 and 10.6019/PXD057629.

GFP pulldowns of HIM-17::GFP for IP/MS was performed in duplicate and corrected against unspecific binding of proteins bound to GFP trap beads. Specifically, potential false positive hits, defined as such because they were also detected in wild type (N2), AIR-2::GFP and AIR-2KD::GFP pulldowns, were removed from this list. Then, only the remaining hits detected on both independent biological repeats, and for which 2 or more peptides were detected in each experiment, made it to this final table. The only exception is KU-80 which made it to the list after subtraction of the unspecific/false positive hits, but for which only 1 peptide was detected in the December pulldown (14 peptides were detected in the Feb 2014 pulldown). Given that we found KU-70 both times and the coverage/depth was better for the Feb 2014 pulldown we decided to keep KU-80 on the final table. Supplemental Table 2 shows all the hits that remained after subtraction of the unspecific binding proteins.

Protein structure predictions

Structure analysis of protein-protein interaction

Prediction of protein-protein interactions is based on available protein structures or AlphaFold2 (AF2) (Jumper et al., 2021 [↗](#)) models. Most relevant proteins DSB-1, DSB-2 and DSB-3 in *C. Elegans* do not have crystal structures. Hence, we rely on AF2 homology models that predicts core domains with confidence factors better than 90, well within the range of applicability to scan for possible protein-protein interactions. Docked poses among these proteins will be predicted using ClusPro (Comeau et al., 2004b [↗](#), 2004a [↗](#); Kozakov et al., 2017 [↗](#)) and free energy complementarity will further analyze with FastContact (Camacho & Zhang, 2005 [↗](#); Champ and Camacho, 2007 [↗](#)), both well-known structural modeling tools developed by one of us.

Peptide design

Peptides were designed using MD simulations in AMBER18 on the GPU-accelerated code with the AMBER ff14SB force field (Salomon-Ferrer et al., 2013 [↗](#); Maier et al., 2015 [↗](#)). The tLeap binary was used to solve structures in an octahedral TIP3P water box with a 15 Å distance from the peptide surface to the box edges and a closeness parameter of 0.75 Å. The system was neutralized and solvated in 150 mM NaCl. The nonbonded interaction cutoff was set to 8 Å. Hydrogen bonds were constrained using the SHAKE algorithm and an integration time step of 2 fs. Simulations were carried out by equilibrating the system for 5 ns at NPT, using a Berendsen thermostat to maintain a constant pressure of 1 atm followed by 300 ns NVT production at 300 K.

Structural Modeling of Full-Length DSB-1, DSB-2, and DSB-3 Interactions

To model the interaction between the full-length DSB-1, DSB-2, and DSB-3 proteins, we used the Mol* Viewer (Molstar), an interactive web-based molecular visualization platform. Predicted structures were imported into the viewer to evaluate their relative orientation, potential interaction surfaces, and spatial compatibility. Full-length models were analyzed using chain-specific coloring and representation tools to facilitate comparative visualization and interpretation of possible interaction interfaces.

Supplementary data

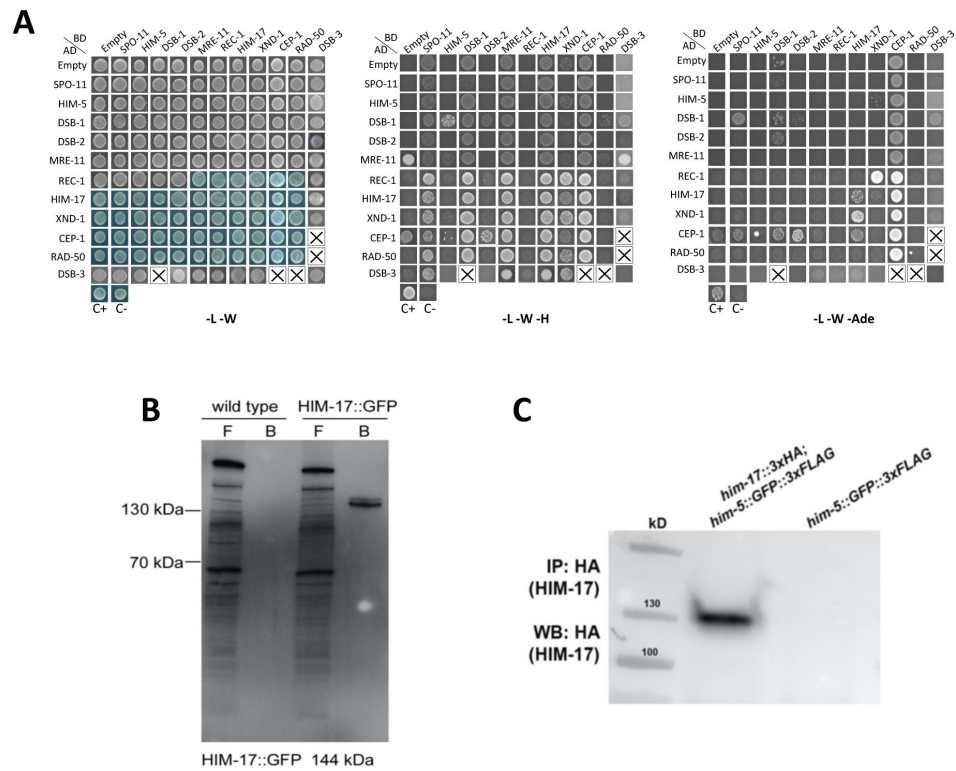


Figure S1

Yeast-2-Hybrid studies and control IP data.

A) Representative Y2H interactions between DSB factors, as monitored by growth in different media: (left) -L-W used as a positive control; (center) -L-W-H showing weak interactions and, in some cases, self-activation; (right) -L-W -Ade showing strong interactions and self-activation by CEP-1. Every DSB factor was tested here as activating domain (AD) fusions as well as reciprocal DNA binding domain (BD) fusions. **B)** Western blot showing immunoprecipitation of HIM-17 from *him-17::GFP* whole worm lysates using anti-GFP antibody. Unbound protein fraction (flow through, F) and proteins bound to the GFP trap beads (**B**) from wild type and *him-17::GFP* transgenic worm lysates are shown. The HIM-17::GFP-specific signal is enriched by using a GFP trap. **C)** Western blot showing immunoprecipitation of HIM-17 from *him-17::3xHA; him-5::GFP::3xFLAG* whole worm lysates using an HA affinity matrix.

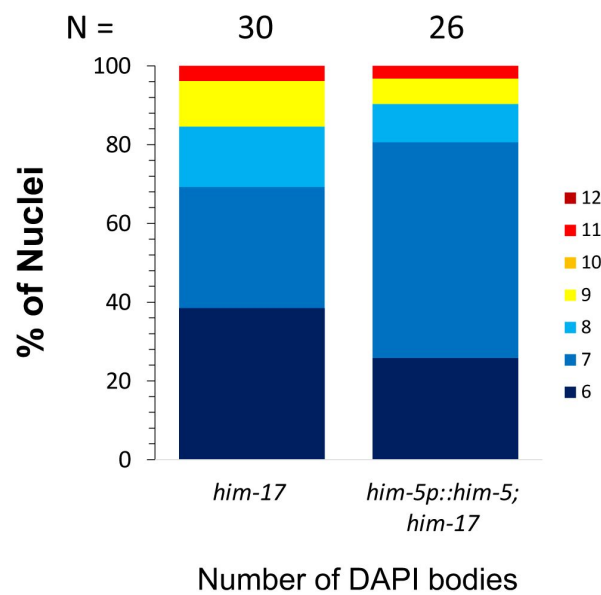


Figure S2

Quantification of DAPI-stained bodies at diakinesis in *him-17(ok424)* shows lack of rescue by HIM-5 expressed from its endogenous promoter.

Colors correspond to the number of DAPI-stained bodies shown in the key. No statistical difference is observed.

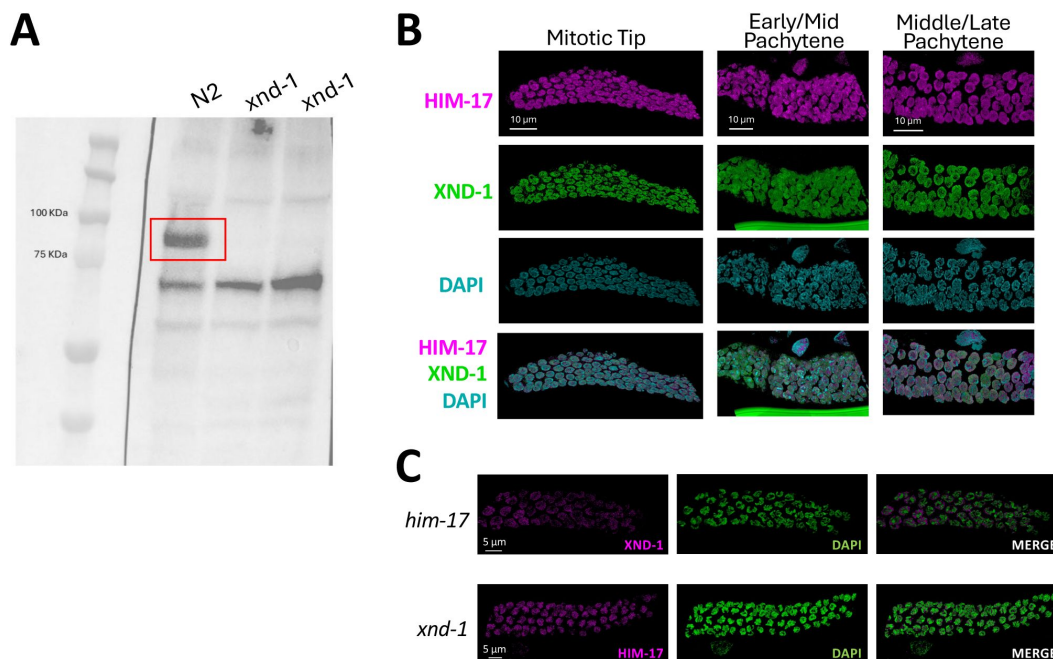


Figure S3

Localization of XND-1 and HIM-17 are non-overlapping and not interdependent.

A) Western blot showing specificity of the guinea-pig anti-XND-1 antibody. Western analysis was performed on protein extracts from wild-type N2 worms and two independent *xnd-1* mutant strains. A distinct band corresponding to XND-1 (red square) is detected only in the N2 extract, confirming the specificity of the antibody. No signal is observed in the mutant strains, consistent with loss of *xnd-1* expression (see also Wagner et al., 2010). **B)** 3xHA::HIM-17 and anti-XND-1 staining do not overlap with one another or with the DNA axes. Shown here are 3D renderings of confocal stacks from the mitotic zone, early-middle pachytene, and mid-late pachytene regions. **C)** Localization of XND-1 is normal in *him-17(ok424^{M-Z})* mutants (anti-XND-1, pink; DNA/DAPI, green) (top). Localization of 3xHA::HIM-17 is unaffected in *xnd-1(ok709)* mutants (anti-HA, pink; DNA/DAPI, green) (bottom).

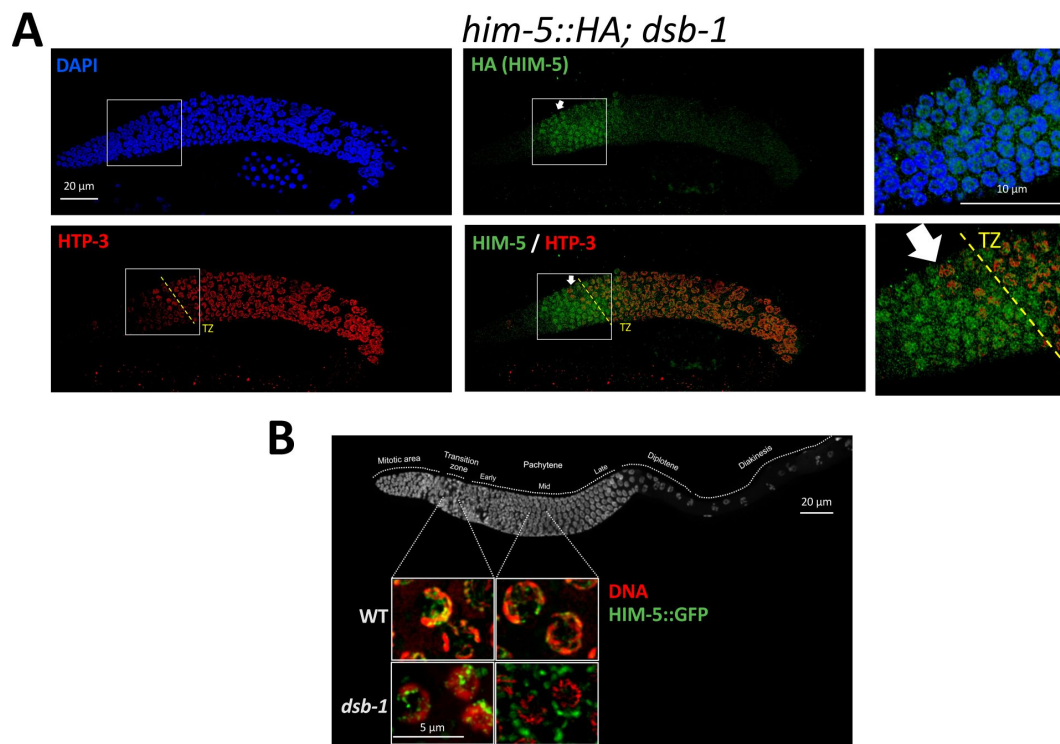


Figure S4.

Localization of endogenous HIM-5 in *dsb-1* mutants.

A) Immunofluorescence analysis of HIM-5::HA in *dsb-1* mutants. DAPI (blue) stains DNA, anti-HA marks endogenously tagged HIM-5 (green), and HTP-3 (red) labels chromosome axes. HIM-5 appears localized in nuclei in pre-meiotic stages (indicated by arrows). However, after the transition zone (TZ), HIM-5 loses its nuclear localization. **B) Top:** *C. elegans* gonad fixed and stained with DAPI to show the organization and distribution of the nuclei along the Prophase I. **Bottom:** live imaging of nuclei in the transition zone (leptotene-zygotene) and middle-pachytene. *eaIs15 (Pie-1::him-5::GFP)* is visualized in freshly dissected gonads by GFP fluorescence (green), and DNA by DRAQ5 (red). In *dsb-1* mutants, HIM-5 is nuclear in the transition zone and then only appears in cytoplasmic puncta by middle pachytene.

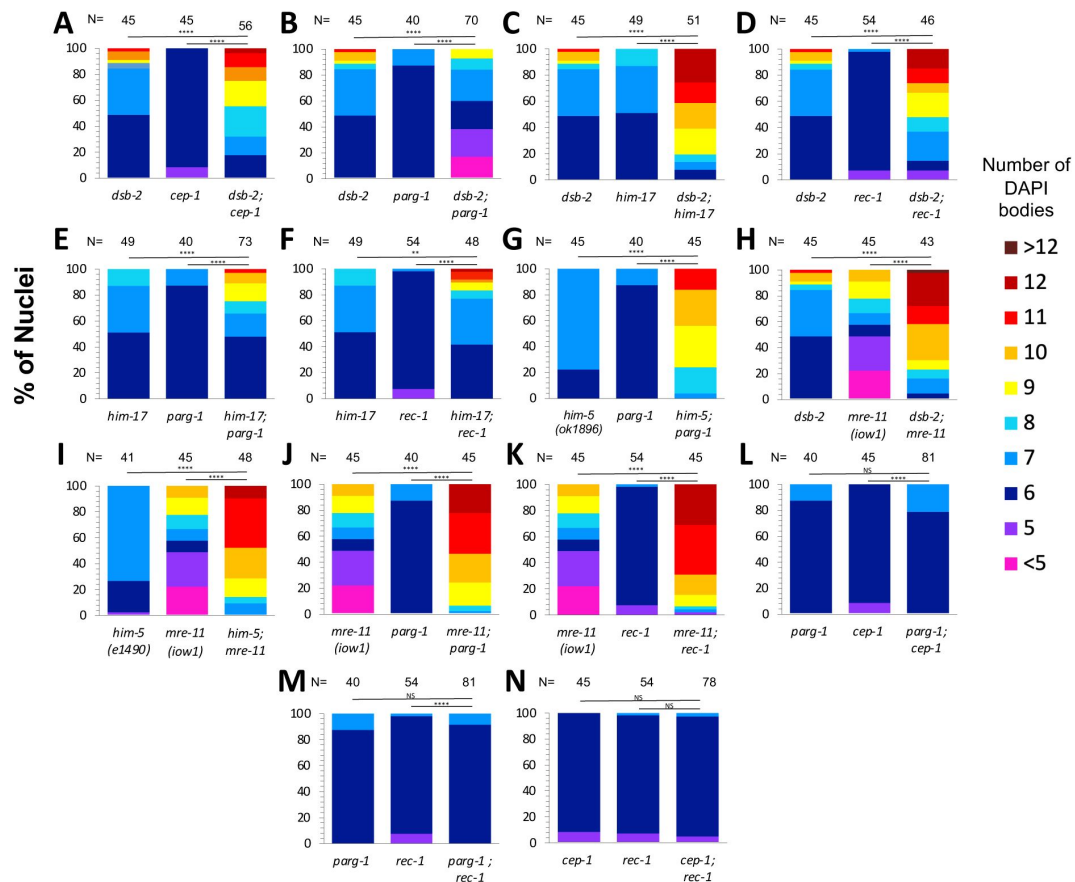


Figure S5.

Epistasis analysis of DSB factors defines multiple genetic groups for crossover formation.

A-N) Crossover formation is assessed by the number of DAPI-staining bodies at diakinesis. Each graph shows the quantification of DAPI-bodies in diakinesis nuclei for the indicated single and double mutants. Color indicates the number of DAPI-staining bodies. Sample sizes (N) are indicated. Statistical significance for comparisons between groups is shown at the top (NS= not significant, **p < 0.01, *** p<0.001, **** p< 0.0001).

Figure S6

Mixed phenotypes are seen in *cep-1(lg12501); mre-11(iow1)* double mutants.

A and E) Quantification of the number of DAPI-stained bodies at diakinesis for the indicated genotypes. Color indicates the number of DAPI-stained bodies. **B-D)** Representative DAPI-stained images of oocytes in diakinesis for *cep-1;mre-11* worms showing univalents (**B, C**) and fusions (**D**).

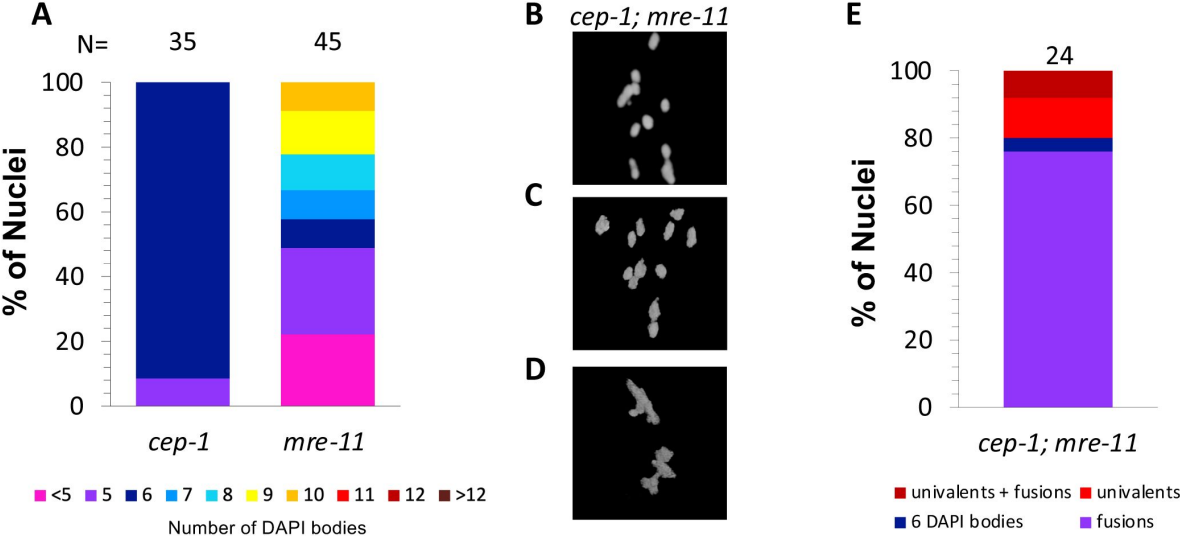
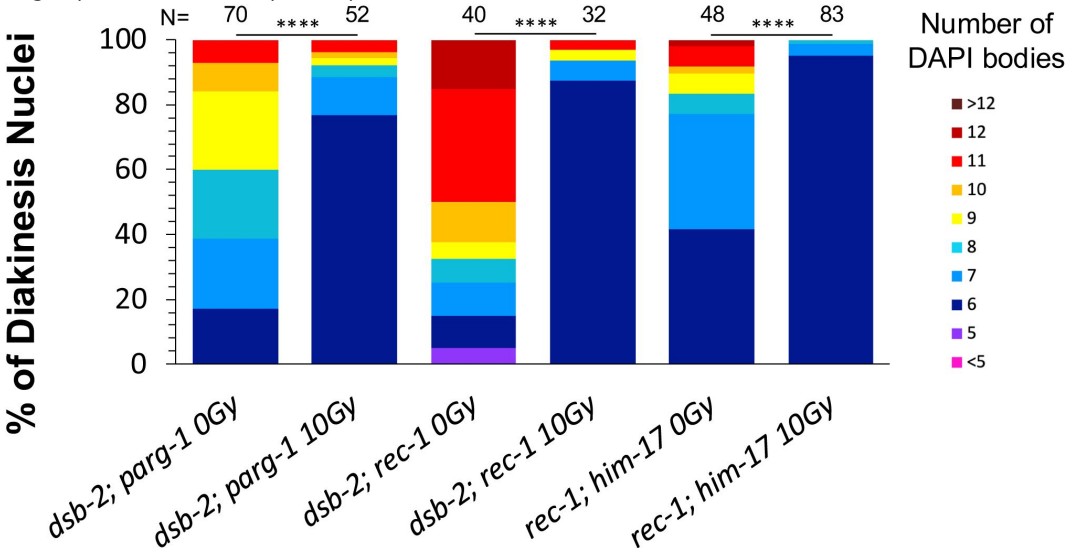


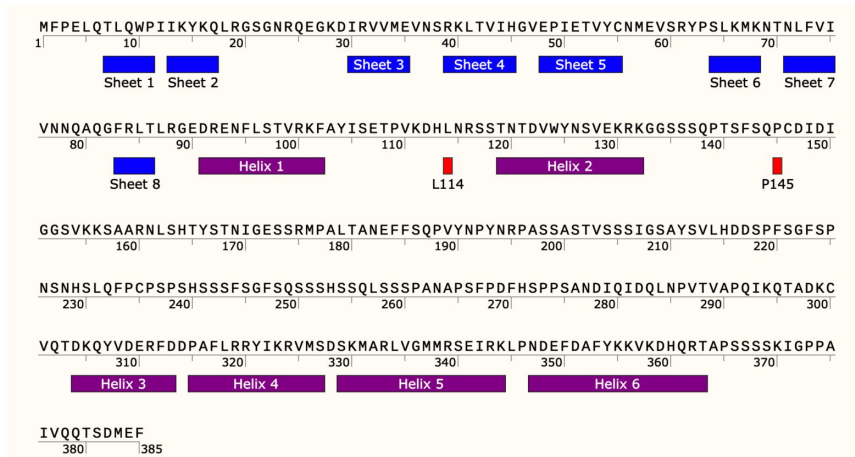
Figure S7

Irradiation rescues crossover defects of accessory factor double mutant strains.

Quantification of DAPI-stained bodies at diakinesis for indicated genotypes with and without 10Gy of γ -irradiation. Color indicates the number of DAPI-staining bodies. Sample sizes (N) are indicated. Statistical significance for comparisons between groups is shown at the top (**** $p < 0.0001$).



A



B

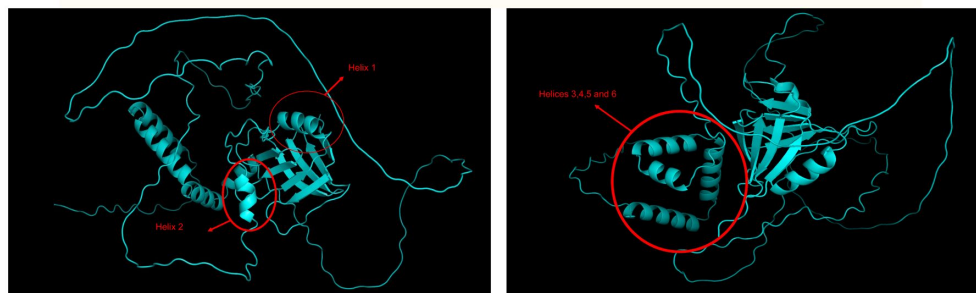


Figure S8

Structural analysis of DSB-1 identified potential interaction motifs.

A) DSB-1 protein sequence with secondary structures demarcated: beta-sheets (blue); alpha-helices (purple). Deletions for Y2H assay shown in **Figure 3** are marked with red and highlight in **B)** Alphafold2 model of DSB-1: helix 2 (left); helices 3-6 (right).

Strain	Genotype	Reference in text
N2	<i>C. elegans</i> var Bristol (N2).	Wild type
AV477	<i>dsb-2(me96)</i> II	<i>dsb-2</i>
AV280	<i>unc-119(e2498)</i> III; <i>him-17(ok424)</i> V; <i>mels5[unc-119(+)+him-17::GFP]</i> .	<i>him-17::GFP</i>
CA1117	<i>dsb-1(we11)</i> IV/ <i>nT1[unc-?(n754) let-?]</i> (IV;V).	<i>dsb-1</i>
CB4088	<i>him-5(e1490)</i> V	<i>him-5(e1490)</i>
CB6036	<i>him-17(e2806)</i> V	<i>him-17</i>
KR5305	<i>rec-1(h2875)</i> I	<i>rec-1</i>
NSV250	<i>him-5(DDR43[him-5::3XHA])</i>	<i>him-5::3xHA</i>
NSV205	<i>him-17(DDR37[him-17::3XHA])</i> V	<i>him-17::3xHA</i>
NSV485	<i>eals4;him-17(DDR37[him-17::3XHA])</i> V	<i>him-17::3xHA;him-5::GFP::3xFLAG</i>
NSV508	<i>gfp::dsb-1;him-5::3xHA</i>	<i>gfp::dsb-1;him-5::3xHA</i>
QP1102	<i>rec-1(h2875)</i> I; <i>dsb-2(me96)</i> II	<i>dsb-2; rec-1</i>
QP1116	<i>mre-11(iow1)</i> , <i>him-5(e1490)</i> V/ <i>nT1 [qls51]</i> ((IV;V)	<i>him-5; mre-11</i>
QP1252	<i>rec-1(h2875);cep-1(lg12501)</i> I	<i>rec-1; cep-1</i>
QP1317	<i>rec-1 (h2875)</i> I; <i>him-17(e2806)/nT1g</i> V	<i>rec-1; him-17</i>
QP1366	<i>parg-1(gk120)</i> IV; <i>him-5(ok1896)</i> V	<i>him-5; parg-1</i>
QP 1367	<i>dsb-2(me96)</i> II; <i>parg-1(gk120)</i> IV	<i>dsb-2; parg-1</i>
QP1368	<i>rec-1(h2875)</i> I; <i>parg-1(gk120)</i> IV	<i>rec-1; parg-1</i>
QP1370	<i>cep-1(lg12501)</i> I; <i>parg-1(gk120)</i> IV	<i>parg-1; cep-1</i>
QP1373	<i>dsb-2(me96)</i> II; <i>him-17(e2806)</i> V	<i>dsb-2; him-17</i>
QP1374	<i>parg-1(gk120)</i> IV; <i>him-17(e2806)</i> V	<i>him-17; parg-1</i>
QP1550	<i>cep-1(lg12501)</i> I; <i>dsb-2(me96)</i> II	<i>cep-1; dsb-2</i>
QP1572	<i>parg-1(gk120)</i> IV; <i>mre-11(iow1)/nT1gU [unc-?(n754) let-?]</i> (IV; V)	<i>mre-11; parg-1</i>
QP1573	<i>cep-1(lg12501)</i> I; <i>mre-11(iow1)/ nT1[qls51]</i> (IV;V)	<i>cep-1; mre-11</i>
QP1623	<i>rec-1(h2875)</i> I; <i>mre-11(iow1)/nT1gU [unc-?(n754) let-?]</i> (IV; V)	<i>mre-11; rec-1</i>
QP1624	<i>dsb-2(me96)</i> II; <i>mre-11(iow1)/ unc-?(n754) let-?</i> (IV; V)	<i>dsb-2; mre-11</i>
QP1710	<i>eals4[Phim-5::him-5::gfp::3xFLAG];him-17(e2707)</i> V	<i>eals4;him-17(e2707)</i>
QP1744	<i>eals4[Phim-5::him-5::gfp::3xFLAG + unc-119(+)];him-17(ok424)</i> V	<i>eals4; him-17(ok424)</i>
QP1749	<i>eals15[Ppie-1::him-5::gfp + unc-119(+)];him-17(ok424)</i> V	<i>eals15;him-17(ok424)</i>
QP1907	<i>dsb-1(we11)</i> IV/ <i>nT1 [qls51]</i> (IV; V); <i>eals15[Ppie-1::him-5::gfp +unc-119(+)]</i>	<i>dsb-1/nT1; eals15</i>
QP1909	<i>dsb-2(me96)</i> II; <i>eals15[Ppie-1::him-5::gfp +unc-119(+)]</i> ;	<i>dsb-2; eals15</i>
QP1961	<i>eals4 (Phim-5::him-5::gfp::3xFLAG::him-5 3' UTR + unc-119(+))</i>	<i>eals4</i>
PCM575	<i>dsb-1(icm97[GFP::dsb-1])</i> IV	<i>GFP::dsb-1</i>
RB869	<i>xnd-1(ok709)</i> III	<i>xnd-1</i>
RB1562	<i>him-5(ok1896)</i> V	<i>him-5(ok1896)</i>
SSM2	<i>mre-11(iow1)/nT1[qls51]</i> (IV;V)	<i>mre-11</i>
VC130	<i>parg-1(gk120)</i> IV	<i>parg-1</i>
VC255	<i>+nT1</i> IV; <i>him-17(ok424)/nT1</i> V	<i>him-17(ok424)</i>
XY1054	<i>cep-1(lg12501)</i>	<i>cep-1</i>

Table S1

Strains and genetics.

All strains were derived from the wild-type Bristol strain N2 and were cultivated at 20 °C under standard conditions. Abbreviated names and full genotypes of the strains used in this study are listed here.

Protein ID	Protein name/ORF	Predicted Molecular Weight (kD)	Sequence Coverage (%)	
			Replicate 1	Replicate 2
CE13489	HIM-17	111.0 (954 aa)	38.7%	42.2%
CE51355	CKU-80	80.7 (713 aa)	1.4%	29.5%
CE28985	MSH-6	133.6 (1186 aa)	8%	12.3%
CE44308	CKU-70	77.9 (679 aa)	13.7%	15.2%
CE30653	RPC-2	129.1 (1154 aa)	7.1%	9.8%
CE42531	F33H1.4	156.5 (1378 aa)	3.6%	4.4%
CE26863	XND-1	78.3 (702 aa)	13.7%	7.4%
CE02015	CID-1	162.6 (1425 aa)	2.3%	5.5%
CE04148*	KLP-10	77.3 (690 aa)	3.6%	7.7%
CE28437	LSY-2	42.3 (365 aa)	10.10%	9.9%
CE41449	MEL-46	110.5 (973 aa)	2.4%	2.5%
CE27382	TAG-153	80.5 (733 aa)	8.7%	8.9%
CE01045	SNFC-5	42.8 (381 aa)	6.8%	6%
CE28922	F26F4.12	25.6 (231 aa)	10.8%	10.8%
CE16631	KLP-19	123.1 (1083 aa)	3.4%	2.1%
CE36164	LSL-1	36.9 (318 aa)	6.9%	7.2%

*Listed as pseudogene in WormBase version: WS288

Table S2

Immunoprecipitation and Mass Spectrometry (IP-MS) Results of HIM-17::GFP from whole worm extracts.

Sequence coverage refers to the percentage of the protein sequence that was pulled down in the HIM-17::GFP IP samples. Numbers are from two independent biological repeats and represent specific enrichment in HIM-17::GFP compared to control IPs (see Methods).

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Author information

Marilina Raices

Magee-Womens Research Institute, Pittsburgh, United States

Fabiola Balmir

Magee-Womens Research Institute, Pittsburgh, United States

Nicola Silva

Department of Biology, Masaryk University, Brno, Czech Republic

ORCID iD: [0000-0001-5406-2280](https://orcid.org/0000-0001-5406-2280)

Wei Li

Magee-Womens Research Institute, Pittsburgh, United States, Tsinghua University Medical School, Beijing, China

McKenzie K Grundy

Department of Microbiology and Molecular Genetics, University of Pittsburgh School of Medicine, UPMC Hillman Cancer Center, Pittsburgh, United States

Dane K Hoffman^{\$}

Magee-Womens Research Institute, Pittsburgh, United States

^{\$}Lester & Sue Smith Breast Center, Department of Molecular & Cellular Biology, Baylor College of Medicine, Houston, TX 77030.

Elisabeth Altendorfer

Department of Genetics, Blavatnik Institute, Harvard Medical School, Boston, United States

Carlos Jaime Camacho

Department of Computational and Systems Biology, University of Pittsburgh School of Medicine, Pittsburgh, United States

ORCID iD: [0000-0003-1741-8529](https://orcid.org/0000-0003-1741-8529)

Kara A Bernstein

Department of Microbiology and Molecular Genetics, University of Pittsburgh School of Medicine, UPMC Hillman Cancer Center, Pittsburgh, United States, Department of Biochemistry and Biophysics, University of Pennsylvania, Penn Center for Genome Integrity, Philadelphia, United States

Monica P Colaiácovo

Department of Genetics, Blavatnik Institute, Harvard Medical School, Boston, United States
ORCID iD: [0000-0001-7803-4372](https://orcid.org/0000-0001-7803-4372)

Judith L Yanowitz

Magee-Womens Research Institute, Pittsburgh, United States, Department of Obstetrics, Gynecology and Reproductive Sciences, University of Pittsburgh School of Medicine, Pittsburgh, United States

ORCID iD: [0000-0001-6886-8787](https://orcid.org/0000-0001-6886-8787)

For correspondence: yanowitzjl@mwri.magee.edu

Editors

Reviewing Editor

Yukiko Yamashita

Whitehead Institute/MIT, Cambridge, United States of America

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Reviewer #1 (Public review):

Summary:

The manuscript by Raices et al., provides some novel insights into the role and interactions between SPO-11 accessory proteins in *C. elegans*. The authors propose a model of meiotic DSBs regulation, critical to our understanding of DSB formation and ultimately crossover regulation and accurate chromosome segregation. The work also emphasizes the commonalities and species-specific aspects of DSB regulation.

Strengths:

This study capitalizes on the strengths of the *C. elegans* system to uncover genetic interactions between a SPO-11 accessory proteins. In combination with physical interactions, the authors synthesize their findings into a model, which will serve as the basis for future work, to determine mechanisms of DSB regulation.

Weaknesses:

The methodology, although standard, still lacks some rigor, especially with the IPs.

<https://doi.org/10.7554/eLife.96458.2.sa3>

Reviewer #2 (Public review):

Summary:

Meiotic recombination initiates with the formation of DNA double-strand break (DSB) formation, catalyzed by the conserved topoisomerase-like enzyme Spo11. Spo11 requires accessory factors that are poorly conserved across eukaryotes. Previous genetic studies have identified several proteins required for DSB formation in *C. elegans* to varying degrees; however, how these proteins interact with each other to recruit the DSB-forming machinery to chromosome axes remains unclear.

In this study, Raices et al. characterized the biochemical and genetic interactions among proteins that are known to promote DSB formation during *C. elegans* meiosis. The authors examined pairwise interactions using yeast two-hybrid (Y2H) and co-immunoprecipitation and revealed an interaction between a chromatin-associated protein HIM-17 and a transcription factor XND-1. They further confirmed the previously known interaction between DSB-1 and SPO-11 and showed that DSB-1 also interacts with a nematode-specific HIM-5, which is essential for DSB formation on the X chromosome. They also assessed genetic interactions among these proteins, categorizing them into four epistasis groups by comparing phenotypes in double vs. single mutants. Combining these results, the authors proposed a model of how these proteins interact with chromatin loops and are recruited to chromosome axes, offering insights into the process in *C. elegans* compared to other organisms.

Weaknesses:

This work relies heavily on Y2H, which is notorious for having high rates of false positives and false negatives. Although the interactions between HIM-17 and XND-1 and between DSB-1 and HIM-5 were validated by co-IP, the significance of these interactions was not tested in vivo. Cataloging Y2H and genetic interactions does not yield much more insight. The model proposed in Figure 4 is also highly speculative.

<https://doi.org/10.7554/eLife.96458.2.sa2>

Reviewer #3 (Public review):

The goal of this work is to understand the regulation of double-strand break formation during meiosis in *C. elegans*. The authors have analyzed physical and genetic interactions among a subset of factors that have been previously implicated in DSB formation or the number of timing of DSBs: CEP-1, DSB-1, DSB-2, DSB-3, HIM-5, HIM-17, MRE-11, REC-1, PARG-1, and XND-1.

The 10 proteins that are analyzed here include a diverse set of factors with different functions, based on prior analyses in many published studies. The term "Spo11 accessory factors" has been used in the meiosis literature to describe proteins that directly promote Spo11 cleavage activity, rather than factors that are important for the expression of meiotic proteins or that influence the genome-wide distribution or timing of DSBs. Based on this definition, the known SPO-11 accessory factors in *C. elegans* include DSB-1, DSB-2, DSB-3, and the MRN complex (at least MRE-11 and RAD-50). These are all homologs of proteins that have been studied biochemically and structurally in other organisms. DSB-1 & DSB-2 are homologs of Rec114, while DSB-3 is a homolog of Mei4. Biochemical and structural studies have shown that Rec114 and Mei4 directly modulate Spo11 activity by recruiting Spo11 to chromatin and promoting its dimerization, which is essential for cleavage. The other factors analyzed in this study affect the timing, distribution, or number of RAD-51 foci, but they likely do so indirectly. As elaborated below, XND-1 and HIM-17 are transcription factors that modulate the expression of other meiotic genes, and their role in DSB formation is parsimoniously explained by this regulatory activity. The roles of HIM-5 and REC-1 remain unclear; the reported localization of HIM-5 to autosomes is consistent with a role in transcription (the autosomes are transcriptionally active in the germline, while the X chromosome is largely silent), but its loss-of-function phenotypes are much more limited than those of HIM-17 and XND-1, so it may play a more direct role in DSB formation. The roles of CEP-1 (a Rad53 homolog) and PARG-1 are also ambiguous, but their homologs in other organisms contribute to DNA repair rather than DSB formation.

An additional significant limitation of the study, as stated in my initial review, is that much of the analysis here relies on cytological visualization of RAD-51 foci as a proxy for DSBs. RAD-51 associates transiently with DSB sites as they undergo repair and is thus limited in its ability to reveal details about the timing or abundance of DSBs since its loading and removal involve additional steps that may be influenced by the factors being analyzed.

The paper focuses extensively on HIM-5, which was previously shown through genetic and cytological analysis to be important for breaks on the X chromosome. The revised manuscript still claims that "HIM-5 mediates interactions with the different accessory factors sub-groups, providing insights into how components on the DNA loops may interact with the chromosome axis." The weak interactions between HIM-5 and DSB-1/2 detected in the Y2H assay do not convincingly support such a role. The idea that HIM-5 directly promotes break formation is also inconsistent with genetic data showing that *him-5* mutants lack breaks on the X chromosomes, while HIM-5 has been shown to be enriched on autosomes. Additionally, as noted in my comment to the authors, the localization data for HIM-5 shown in this paper are discordant with prior studies; this discrepancy should be addressed experimentally.

This paper describes REC-1 and HIM-5 as paralogs, based on prior analysis in a paper that included some of the same authors (Chung et al., 2015; DOI 10.1101/gad.266056.115). In my initial review I mentioned that this earlier conclusion was likely incorrect and should not be propagated uncritically here. Since the authors have rebutted this comment rather than amending it, I feel it is important to explain my concerns about the conclusions of previous study. Chung et al. found a small region of potential homology between the *C. elegans* *rec-1* and *him-5* genes and also reported that *him-5*; *rec-1* double mutants have more severe

defects than either single mutant, indicative of a stronger reduction in DSBs. Based on these observations and an additional argument based on microsynteny, they concluded that these two genes arose through recent duplication and divergence. However, as they noted, genes resembling *rec-1* are absent from all other *Caenorhabditis* species, even those most closely related to *C. elegans*. The hypothesis that two genes are paralogs that arose through duplication and divergence is thus based on their presence in a single species, in the absence of extensive homology or evidence for conserved molecular function. Further, the hypothesis that gene duplication and divergence has given rise to two paralogs that share no evident structural similarity or common interaction partners in the few million years since *C. elegans* diverged from its closest known relatives is implausible. In contrast, DSB-1 and DSB-2 are both homologs of *Rec114* that clearly arose through duplication and divergence within the *Caenorhabditis* lineage, but much earlier than the proposed split between *REC-1* and *HIM-5*. Two genes that can be unambiguously identified as *dsb-1* and *dsb-2* are present in genomes throughout the *Elegans* supergroup and absent in the *Angaria* supergroup, placing the duplication event at around 18-30 MYA, yet DSB-1 and DSB-2 share much greater similarity in their amino acid sequence, predicted structure, and function than *HIM-5* and *REC-1*. Further, Raices place *HIM-5* and *REC-1* in different functional complexes (Figure 3B).

The authors acknowledge that *HIM-17* is a transcription factor that regulates many meiotic genes. Like *HIM-17*, *XND-1* is cytologically enriched along the autosomes in germline nuclei, suggestive of a role in transcription. The Reinke lab performed ChIP-seq in a strain expressing an *XND-1::GFP* fusion protein and showed that it binds to promoter regions, many of which overlap with the *HIM-17*-regulated promoters characterized by the Ahringer lab (doi: 10.1126/sciadv.abo4082). Work from the Yanowitz lab has shown that *XND-1* influences the transcription of many other genes involved in meiosis (doi: 10.1534/g3.116.035725) and work from the Colaiacovo lab has shown that *XND-1* regulates the expression of *CRA-1* (doi: 10.1371/journal.pgen.1005029). Additionally, loss of *HIM-17* or *XND-1* causes pleiotropic phenotypes, consistent with a broad role in gene regulation. Collectively, these data indicate that *XND-1* and *HIM-17* are transcription factors that are important for the proper expression of many germline-expressed genes. Thus, as stated above, the roles of *HIM-17* and *XND-1* in DSB formation, as well as their effects on histone modification, are parsimoniously explained by their regulation of the expression of factors that contribute more directly to DSB formation and chromatin modification. I feel strongly that transcription factors should not be described as "SPO-11 accessory factors."

<https://doi.org/10.7554/eLife.96458.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The manuscript by Raices et al., provides novel insights into the role and interactions between SPO-11 accessory proteins in C. elegans. The authors propose a model of meiotic DSBs regulation, critical to our understanding of DSB formation and ultimately crossover regulation and accurate chromosome segregation. The work also emphasizes the commonalities and species-specific aspects of DSB regulation.

Strengths:

*This study capitalizes on the strengths of the *C. elegans* system to uncover genetic interactions between a large number of SPO-11 accessory proteins. In combination with physical interactions, the authors synthesize their findings into a model, which will serve as the basis for future work, to determine mechanisms of DSB regulation.*

Weaknesses:

The methodology, although standard, lacks quantification. This includes the mass spectrometry data, along with the cytology. The work would also benefit from clarifying the role of the DSB machinery on the X chromosome versus the autosomes.

- We have uploaded the MS data and added a summary table with the number of peptides and coverage.
- We have added statistics to the comparisons of DAPI body counts.
- We have provided additional images of the change in HIM-5 localization
- We have quantified the overlap (or lack thereof) between XND-1 and HIM-17 and the DNA axis

Reviewer #2 (Public Review):

Summary:

*Meiotic recombination initiates with the formation of DNA double-strand break (DSB) formation, catalyzed by the conserved topoisomerase-like enzyme Spo11. Spo11 requires accessory factors that are poorly conserved across eukaryotes. Previous genetic studies have identified several proteins required for DSB formation in *C. elegans* to varying degrees; however, how these proteins interact with each other to recruit the DSB-forming machinery to chromosome axes remains unclear.*

*In this study, Raices et al. characterized the biochemical and genetic interactions among proteins that are known to promote DSB formation during *C. elegans* meiosis. The authors examined pairwise interactions using yeast two-hybrid (Y2H) and co-immunoprecipitation and revealed an interaction between a chromatin-associated protein HIM-17 and a transcription factor XND-1. They further confirmed the previously known interaction between DSB-1 and SPO-11 and showed that DSB-1 also interacts with a nematodespecific HIM-5, which is essential for DSB formation on the X chromosome. They also assessed genetic interactions among these proteins, categorizing them into four epistasis groups by comparing phenotypes in double vs. single mutants. Combining these results, the authors proposed a model of how these proteins interact with chromatin loops and are recruited to chromosome axes, offering insights into the process in *C. elegans* compared to other organisms.*

Weaknesses:

This work relies heavily on Y2H, which is notorious for having high rates of false positives and false negatives. Although the interactions between HIM-17 and XND-1 and between DSB-1 and HIM-5 were validated by co-IP, the significance of these interactions was not tested, and cataloging Y2H interactions does not yield much more insight.

We appreciate that the reviewer recognized the value of our IP data, but we beg to differ that we rely too heavily on the Y2H. We also provide genetic analysis on bivalent formation to support the physical interaction data. We do acknowledge that there are caveats with Y2H, however, including that a subset of the interactions can only be examined with proteins in

one orientation due to auto-activation. While we acknowledge that it would be nice to have IP data for all of the proteins using CRISPR-tagged, functional alleles, these strains are not all feasible (e.g. no functional *rec-1* tag has been made) and are beyond the scope of the current work.

Moreover, most experiments lack rigor, which raises serious concerns about whether the data convincingly supports the conclusions of this paper. For instance, the XND-1 antibody appears to detect a band in the control IP; however, there was no mention of the specificity of this antibody.

We previously showed the specificity of this antibody in its original publication showing lack of staining in the *xnd-1* mutant by IF (Wagner et al., 2010). To further address this, however, we have now included a new supplementary figure (Figure S1) demonstrating the specificity of the XND-1 antibody by Western blot. The antibody detects a distinct band in extracts from wild-type (N2) worms, but this band is absent in two independent *xnd-1* mutant strains. This confirms that the antibody specifically recognizes XND-1, supporting the validity of the IP results shown in the main figures.

Additionally, epistasis analysis of various genetic mutants is based on the quantification of DAPI bodies in diakinesis oocytes, but the comparisons were made without statistical analyses.

We have added statistical analysis to all datasets where quantification was possible, strengthening the rigor and interpretation of our findings.

*For cytological data, a single representative nucleus was shown without quantification and rigorous analysis. The rationale for some experiments is also questionable (e.g. the rescue by *dsb-2* mutants by *him-5* transgenes in Figure 2), making the interpretation of the data unclear. Overall, while this paper claims to present "the first comprehensive model of DSB regulation in a metazoan", cataloging Y2H and genetic interactions did not yield any new insights into DSB formation without rigorous testing of their significance in vivo. The model proposed in Figure 4 is also highly speculative.*

Regarding the cytology, we provide new images and quantification of HIM-17 and XND-1 overlap with the DNA axes. We also added full germ line images showing HIM-5 localization in wild type and *dsb-1* mutants, to provide a more complete and representative view of the observed phenotype. To further support our findings, we've also included images demonstrating that this phenotype is consistently observed with both in live worm with the *him-5::GFP* transgene and in fixed worms with an endogenously tagged version of HIM-5.

Reviewer #3 (Public Review):

During meiosis in sexually reproducing organisms, double-strand breaks are induced by a topoisomerase-related enzyme, Spo11, which is essential for homologous recombination, which in turn is required for accurate chromosome segregation. Additional factors control the number and genome-wide distribution of breaks, but the mechanisms that determine both the frequency and preferred location of meiotic DSBs remain only partially understood in any organism.

The manuscript presents a variety of different analyses that include variable subsets of putative DSB factors. It would be much easier to follow if the analyses had been more systematically applied. It is perplexing that several factors known to be essential for DSB formation (e.g., cohesins, HORMA proteins) are excluded from this analysis, while it includes several others that probably do not directly contribute to DSB formation (XND-1, HIM-17, CEP-1, and PARG-1).

We respectfully disagree with the reviewer's statement regarding the selection of factors included in our analysis. In this work, our focus was specifically on SPO-11 accessory factors — proteins that directly interact with or regulate SPO-11 activity during double-strand break formation. Cohesins and chromosome axis proteins (such as the HORMA domain proteins) are essential for establishing the correct chromosome architecture that supports DSB formation, but there is no evidence that they are direct accessory factors of SPO-11. Therefore, they were intentionally excluded from this study to maintain a clear and focused scope on proteins that more directly modulate SPO-11 function.

Conversely, XND-1, HIM-17, CEP-1, and PARG-1 have all been implicated in regulating aspects of SPO-11-mediated DSB formation or its immediate environment. Although their contributions may involve broader chromatin or DNA damage response regulation, prior literature supports their inclusion as relevant modulators of SPO-11 activity, justifying their analysis within the context of this work.

The strongest claims seem to be that "HIM-5 is the determinant of X-chromosome-specific crossovers" and "HIM-5 coordinates the actions of the different accessory factors subgroups." Prior work had already shown that mutations in him-5 preferentially reduce meiotic DSBs on the X chromosome. While it is possible that HIM-5 plays a direct role in DSB induction on the X chromosome, the evidence presented here does not strongly support this conclusion. It is also difficult to reconcile this idea with evidence from prior studies that him-5 mutations predominantly prevent DSB formation on the sex chromosomes, while the protein localizes to autosomes.

HIM-5 is not the only protein that is autosomally enriched but preferentially affects the X chromosome: MES-4 and MRG-1 are both autosomally-enriched but influence silencing of the X chromosome. While HIM-5 appears autosomally-enriched, it does not appear to be autosomal-exclusive. While we would ideally perform ChIP to determine its localization on chromatin, this method for assaying DSB sites is likely insufficient to identify DSB sites which differ in each nucleus and for which there are no known hotspots in the worm.

him-5 mutants confer an ~50% reduction in total number of breaks and a very profound change in break dynamics (seen by RAD-51 foci (Meneely et al., 2012)). Since the autosomes receives sufficient breaks in this context to attain a crossover in >98% of nuclei, this indicates that the autosomes are much less profoundly impacted by loss of DSB functions than is the X chromosome. Indeed, prior data from co-author, Monica Colaiacovo, showed that fewer breaks occur on the X (Gao, 2015) likely resulting from differences in the chromatin composition of the X and autosome resulting from X chromosome silencing.

The conclusion that HIM-5 must be required for breaks on the X comes from the examination of DSB levels and their localization in different mutants that impair but do not completely abrogate breaks. In any situation where HIM-5 protein expression is affected (xnd-1, him-17, and him-5 null alleles), breaks on the X are reduced/ eliminated. By contrast, in dsb-2 mutants, where HIM-5 expression is unaffected, both X and autosomal breaks are impacted equally. As discussed above, in the absence of HIM-5 function, there are ~15 breaks/ nucleus. The Ppie1::him-5 transgene is expressed to lower levels than Phim-5::him-5, but in the best case, the ectopic expression of this protein should give a maximum of ~15 breaks (the total # of breaks is thought to be ~30/nucleus). By these estimates, Ppie-1::him-5; him-17 and him-5 null mutants have the same number of breaks. Yet, in the former case, breaks occur on the X; whereas in the latter they do not. The best explanation for this discrepancy is that HIM-5 is sufficient to recruits the DSB machinery to the X chromosome.

The one experiment that seems to elicit the conclusion that HIM-5 expression is sufficient for breaks on the X chromosome is flawed (see below). The conclusion that HIM-5

"coordinates the activities of the different accessory sub-groups" is not supported by data presented here or elsewhere.

We have reorganized the discussion to more directly address the reviewers' concerns. We raise the possibility that HIM-5 has an important role in bringing together the SPO-11 and its interacting components (DSB-1/2/3) with the other DSB inducing factors, including those factors that regulating DSB timing (XND-1), coordination with the cell cycle (REC-1), association with the chromosome axis (PARG-1, MRE-11), and coupling to downstream resection and repair (MRE-11, CEP-1).

This raises a natural question: if HIM-5 has such a central role, why are the phenotypes of HIM-5 so mild? We propose that while the loss of DSBs on the X appears mild, more profound effects are seen in the total number, timing, and placement of the DSBs across the genome- all of which are diminished or altered in the absence of HIM-5. The phenotypes of him-5 loss reminiscent of those observed in Prdm9^{-/-} in mice where breaks are relocated to transcriptional start sites and show significant delay in formation. As with PRDM9, the comparatively subtle phenotypes of HIM-5 loss do not diminish its critical role in promoting proper DSB formation in most mammals.

Like most other studies that have examined DSB formation in C. elegans, this work relies on indirect assays, here limited to the cytological appearance of RAD-51 foci and bivalent chromosomes, as evidence of break formation or lack thereof. Unfortunately, neither of these assays has the power to reveal the genome-wide distribution or number of breaks. These assays have additional caveats, due to the fact that RAD-51 association with recombination intermediates and successful crossover formation both require multiple steps downstream of DSB induction, some of which are likely impaired in some of the mutants analyzed here. This severely limits the conclusions that can be drawn. Given that the goal of the work is to understand the effects of individual factors on DSB induction, direct physical assays for DSBs should be applied; many such assays have been developed and used successfully in other organisms.

We appreciate the reviewer's thoughtful comments. We agree that RAD-51 foci are an indirect readout of DSB formation and that their dynamics can be influenced by defects in downstream repair processes. However, in C. elegans, the available methods for directly detecting DSBs are limited. Unlike other organisms, C. elegans lacks γH2AX, eliminating the possibility of using γH2AX as a DSB marker. TUNEL assays, while conceptually appealing, have proven unreliable and poorly reproducible in the germline context. Similarly, RPA foci do not consistently correlate with the number of DSBs and are influenced by additional processing steps.

Given these limitations, RAD-51 foci remain the most widely accepted surrogate for monitoring DSB formation in C. elegans. While we fully acknowledge the caveats associated with this approach — particularly the potential effects of downstream repair defects — RAD-51 analysis continues to provide valuable insight into DSB dynamics and regulation, especially when interpreted in combination with other phenotypic assessments.

Throughout the manuscript, the writing conflates the roles played by different factors that affect DSB formation in very different ways. XND-1 and HIM-17 have previously been shown to be transcription factors that promote the expression of many germline genes, including genes encoding proteins that directly promote DSBs. Mutations in either xnd-1 or him-17 result in dysregulation of germline gene expression and pleiotropic defects in meiosis and fertility, including changes in chromatin structure, dysregulation of meiotic progression, and (for xnd-1) progressive loss of germline immortality. It is thus misleading to refer to HIM-17 and XND-1 as DSB "accessory factors" or to lump their

activities with those of other proteins that are likely to play more direct roles in DSB induction.

It is clear that we will not reach agreement about the direct vs indirect roles here of chromatin remodelers/transcription factors in break formation. In yeast, there is a precedent for SPP1 and in mouse for Prdm9, both of which could be described as transcription factors as well, as having roles in break formation by creating an open chromatin environment for the break machinery. We envision that these proteins function in the same fashion. The changes in histone acetylation in the *xnd-1* mutants supports such a claim.

We do not know what the reviewer is referring to in statement that “XND-1 and HIM-17 have previously been shown to be transcription factors that promote the expression of many germline genes.” While the Carelli et al paper indeed shows a role for HIM-17 in expression of many germline genes, there is only one reference to XND-1 in this manuscript (Figure S3A) which shows that half of XND-1 binding sites overlap with the co-opted germline promoters. There is no transcriptional data at all on *xnd-1* mutants, save our studies (referenced herein) that XND-1 regulates *him-5* expression.

For example, statements such as the following sentence in the Introduction should be omitted or explained more clearly: “xnd-1 is also unique among the accessory factors in influencing the timing of DSBs; in the absence of xnd-1, there is precocious and rapid accumulation of DSBs as monitored by the accumulation of the HR strand-exchange protein RAD-51.

We are not sure what is confusing here. The distribution of RAD-51 foci is significantly altered in *xnd-1* mutants and peak levels of breaks are achieved as nuclei leave the transition zone (Wagner et al., 2010; McClendon et al., 2016). There is no other mutation that causes this type of change in RAD-51 distribution.

“The evidence that HIM-17 promotes the expression of him-5 presented here corroborates data from other publications, notably the recent work of Carelli et al. (2022), but this conclusion should not be presented as novel here.

We have clarified this in the text. We note that this paper showed alterations in *him-5* levels by RNA-Seq but they did not validate these results with quantitative RT-PCR. Thus, our studies do provide an important validation of their prior results.

The other factors also fall into several different functional classes, some of which are relatively well understood, based largely on studies in other organisms. The roles of RAD50 and MRE-11 in DSB induction have been investigated in yeast and other organisms as well as in several prior studies in C. elegans. DSB-1, DSB-2, and DSB-3 are homologs of relatively well-studied meiotic proteins in other organisms (Rec114 and Mei4) that directly promote the activity of Spo11, although the mechanism by which they do so is still unclear.

Whilst we agree that we understand some of the functions of the homologs, there are clearly examples in other processes of conserved proteins adopting unique regulatory function. We should not presume evolutionary conservation until proven. Indeed the comparison between the Mer2 proteins becomes particularly relevant here. For example, the RMM complex in plants does not contain PRD3, although this protein is thought to have function in DSB formation and repair (Lambing et al, 2022; Vrielynck et al., 2021; Thangavel et al., 2023). In *Sordaria*, as well, the Mer2 homolog has distinct functions (Tesse et al., 2017).

Mutations in PARG-1 (a Poly-ADP ribose glycohydrolase) likely affect the regulation of polyADP-ribose addition and removal at sites of DSBs, which in turn are thought to regulate chromatin structure and recruitment of repair factors; however, there is no convincing evidence that PARG-1 directly affects break formation.

Our prior collaborative studies on PARG-1 showed that it has a non-catalytic function that promotes DSBs that is independent of accumulation of PAR (Janisiw et al., 2020; Trivedi et al., 2022)

CEP-1 is a homolog of p53 and is involved in the DNA damage response in the germline, but again is unlikely to directly contribute to DSB induction.

We respectfully disagree with the reviewer's statement. While CEP-1 is indeed a homolog of p53 and plays a major role in the DNA damage response, prior work from Brent Derry's lab and from our group (Mateo et al., 2016) demonstrated that specific cep-1 separation-of-function alleles affect DSB induction and/or repair pathway choice independently of canonical DNA damage checkpoint activation. In particular, defects in DSB formation observed in certain cep-1 mutants can be rescued by exogenous irradiation, supporting a direct or closely linked role in promoting DSB formation rather than merely responding to damage. Thus, based on these functional data, we considered CEP-1 a relevant factor to include in our analysis. We have now clarified this rationale in the revised manuscript.

HIM-5 and REC-1 do not have apparent homologs in other organisms and play poorly understood roles in promoting DSB induction. A mechanistic understanding of their functions would be of value to the field, but the current work does not shed light on this. A previous paper (Chung et al. G&D 2015) concluded that HIM-5 and REC-1 are paralogs arising from a recent gene duplication, based on genetic evidence for a partially overlapping role in DSB induction, as well as an argument based on the genomic location of these genes in different species; however, these proteins lack any detectable sequence homology and their predicted structures are also dissimilar (both are largely unstructured but REC-1 contains a predicted helical bundle lacking in HIM-5). Moreover, the data presented here do not reveal overlapping sets of genetic or physical interactions for the two genes/proteins. Thus, this earlier conclusion was likely incorrect, and this idea should not be restated uncritically here or used as a basis to interpret phenotypes.

Actually, there is quite good bioinformatic analysis that the rec-1 and him-5 loci evolved from a gene duplication and that each share features of the ancestral protein (Chung et al., 2015). We are sorry if the reviewer casts aspersions on the prior literature and analyses. The homology between these genes with the ancestral protein is near the same degree as dsb-1, dsb-2, or dsb-3 to their ancestral homologs (<17%).

DSB-1 was previously reported to be strictly required for all DSB and CO formation in C. elegans. Here the authors test whether the expression of HIM-5 from the pie-1 promoter can rescue DSB formation in dsb-1 mutants, and claim to see some rescue, based on an increase in the number of nuclei with one apparent bivalent (Figure 2C). This result seems to be the basis for the claim that HIM-5 coordinates the activities of other DSB proteins. However, this assay is not informative, and the conclusion is almost certainly incorrect. Notably, a substantial number of nuclei in the dsb-1 mutant (without Ppie-1::him-5) are reported as displaying a single bivalent (11 DAPI staining bodies) despite prior evidence that DSBs are absent in dsb-1 mutants; this suggests that the way the assay was performed resulted in false positives (bivalents that are not actually bivalents), likely due to inclusion of nuclei in which univalents could not be unambiguously resolved in the microscope. A slightly higher level of nuclei with a single unresolved pair of

chromosomes in the dsb-1; Ppie-1::him-5 strain is thus not convincing evidence for rescue of DSBs/CO formation, and no evidence is presented that these putative COs are X-specific. The authors should provide additional experimental evidence - e.g., detection of RAD-51 and/or COSA-1 foci or genetic evidence of recombination - or remove this claim. The evidence that expression of Ppie-1::him-5 may partially rescue DSB abundance in dsb-2 mutants is hard to interpret since it is currently unknown why C. elegans expresses 2 paralogs of Rec114 (DSB-1 and DSB-2), and the age-dependent reduction of DSBs in dsb-2 mutants is not understood.

We have removed this claim in part because we have been unable to create the triple mutants strains to analyze COSA-1 foci.

To the point about 11 vs 12 DAPI bodies: the literature is actually replete with examples of 11 DAPI bodies vs 12 in mutants with no breaks:

Hinman al., 2021: null allele of dsb-3 has an average of 11.6 +/- 0.6 breaks;

Stamper et al, 2013, show just over 60% of dsb-1 nuclei with 12 DAPI bodies and 5-10% with 10 DAPI bodies. (Figure 1);

In addition, we also previously showed (Machovina et al., 2016) that a subset of meiotic nuclei have a single RAD-51 focus and can achieve a crossover. RAD-51 foci in spo-11 were also reported in Colaiacovo et al., 2003.

Several of the factors analyzed here, including XND-1, HIM-17, HIM-5, DSB-1, DSB-2, and DSB-3, have been shown to localize broadly to chromatin in meiotic cells. Coimmunoprecipitation of pairs of these factors, even following benzonase digestion, is not strong evidence to support a direct physical interaction between proteins.

Similarly, the super-resolution analysis of XND-1 and HIM-17 (Figure 1EF) does not reveal whether these proteins physically interact with each other, and does not add to our understanding of these proteins functions, since they are already known to bind to many of the same promoters. Promoters are also likely to be located in chromatin loops away from the chromosome axis, so in this respect, the localization data are also confirmatory rather than novel.

While the binding to promoters would be expected to be on DNA loops, that has not been definitively shown in the worm germ line. The supplemental data of the Carelli paper suggests that there are ~250 binding sites for each protein at these coopted promoters. This could not account for crossover map seen in C. elegans.

The reviewer states correct that we do not reveal that these proteins interact, but we have shown that the two proteins co-IP and have a Y2H interaction. This interaction is supported by a recent publication (Blazickova et al., 2025) corroborating this conclusion and identifies XND-1 in HIM-17 co-IPs also in the presence of benzonase. We do now show, however, by immuno-localization that the two proteins appear to be adjacent, but nonoverlapping. As now described in the text, AlphaFold 3 modeling and structural analysis suggests that the two proteins do interact directly and that the tagged 5' end of HIM-17 used in our studies is likely to be at least 200nm from the putative XND-1 binding interface, a distance that is consistent with our confocal images showing frequent juxtaposition of the two proteins.

The phenotypic analysis of double mutant combinations does not seem informative. A major problem is that these different strains were only assayed for bivalent formation, which (as mentioned above) requires several steps downstream of DSB induction. Additionally, the basis for many of the single mutant phenotypes is not well understood, making it particularly challenging to interpret the effects of double mutants. Further,

some of the interactions described as "synergistic" appear to be additive, not synergistic. While additive effects can be used as evidence that two genes work in different pathways, this can also be very misleading, especially when the function of individual proteins is unknown. I find that the classification of genes into "epistasis groups" based on this analysis does not shed light on their functions and indeed seems in some cases to contradict what is known about their functions. '

As described above, each of the proteins analyzed is thought to have a direct role in regulating meiotic DSB formation and single mutant phenotypes are consistent with this interpretation. In almost all-if not all- of these cases, IR induced breaks suppress univalent phenotypes (or uncover a downstream repair defect (e.g. in mre-11)) supporting this conclusion. We have changed the terminology from "epistasis groups" since this is not strict epistasis, but rather, "functional groups".

The yeast two-hybrid (Y2H) data are only presented as a single colony. While it is understandable to use a 'representative' colony, it is ideal to include a dilution series for the various interactions, which is how Y2H data are typically shown.

The Y2H data are presented as spots on a plate and are from three to four individual transformants per interaction tested, and are not individual colonies. The experiment was repeated in triplicate from different transformations. We have now made this clearer in the materials and methods section. This approach has been successfully used to examine protein interactions in our prior manuscripts of yeast and human proteins [Gaines et al (2015) Nat. Comms 6:7834; Kondrashova et al (2017) Cancer Discovery 7:984; Garcin et al (2019) PLoS Genetics 15:e1008355; Bonilla et al (2021) eLife 1: e68080) Prakash et al (2022) PNAS 119: e2202727119, etc]

Additional (relatively minor) concerns about these data:

(1) Several interactions reported here seem to be detected in only one direction - e.g., MRE-11-AD/HIM-5-BD, REC-1-AD/XND-1-BD, and XND-1-AD/HIM-17-BD - while no interactions are seen with the reciprocal pairs of fusion proteins. I'm not sure if some of this is due to pasting "positive" colony images into the wrong position in the grid, but this should be addressed.

The asymmetry in the interactions observed is due to the well-known phenomenon in yeast two-hybrid (Y2H) assays where certain plasmids exhibit self-activation when fused in one orientation, making interpretation of reciprocal interactions challenging. In our experiment, some of the plasmids indeed showed self-activation in one direction, which likely accounts for the lack of interaction seen with the reciprocal pairs of fusion proteins. We have clarified this point in the Methods.

(2) DSB-3 was only assayed in pairwise combinations with a subset of other proteins; this should be explained; it is also unclear why the interaction grids are not symmetrical about the diagonal.

We have now completed the analysis by adding the interactions of DSB-3 with the remaining proteins that were missing from the initial set.

(3) I don't understand why the graphic summaries of Y2H data are split among 3 different figures (1, 2, and 3).

We chose to split the graphic summaries of the Y2H data across Figures 1, 2, and 3 because we felt this organization better aligns with the flow of the results presented in each figure. Each

set of interactions is shown in the context of the specific experiments and findings discussed in those sections, which we believe helps provide a clearer and more logical presentation of the data.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Figure 1: B) The IP is difficult to interpret - there is a band of the corresponding size to XND-1 in the control lane calling into question the specificity of the IP/Western.

We added a supplemental figure with the specificity of the antibody showing that there is a background non-specific band.

C) More information about the mass spectrometry should be included. No indication of the number of times a peptide was identified, or the overall coverage of the identified proteins.

Done

This is important as in the results section (line 114) the authors indicate that there was "strong" interaction yet there is no way to assess this.

D) Why wasn't hatching measured in the him-5p::him-5; him-17(ok424) strain?

Great question. I guess we need to do this while back out for review. If anyone has suggestions of what to say here. Clearly we overlooked this point but do have the strain.

E) Quantification of the cytology should be included.

We have now quantified overlap between XND-1 and HIM-17

Figure 2: C) Statistics should be included.

Done

E) Quantification should be included for the cytology. I recommend changing the eals15 to HIM-5.

We included better images showing whole gonads instead of one or two nuclei. We were not sure what the reviewers want us to quantify here since the relocalization of the protein to the cytoplasm is very clear.

I have a general issue with the use of the term epistasis - this is used to order gene function based on different mutant phenotypes, usually with null alleles. While I think the authors have valid points with how they group the different SPO-11 accessory proteins, I do not think they should use the word epistasis, but rather genetic interactions.

We appreciate the reviewers thoughts on this matter and have removed the term epistasis and use functional groups or genetic interactions throughout the text.

Figure 4 and the nature of the X chromosome: First, I think it would help the non-C. elegans reader to include a little more information on the X chromosome with respect to its differences compared to the autosomes. I also think that, if possible, it would be beneficial to include a model of the X in Figure 4.

We have added more about X/autosome differences in the intro and during the discussion of HIM-5 function and have added a figure showing difference in the behavior of the X/autosomes during DSB/crossover formation.

Minor points:

Abstract: Given the findings of Silva and Smolikove on SPO-11 breaks, I recommend removing "early" from line 28 in the Abstract.

Done

Introduction (line 93): I think "biochemical studies" is a stretch here - I recommend "interaction studies".

Done

Results: (lines 160-161): mutations are not required for breaks. Line 172, there is a problem with the sentence.

Corrected

Reviewer #2 (Recommendations For The Authors):

Major comments:

(1) Figure 1B- The signal for XND-1 seems to appear both in the control and him-17::HA IP. Do the authors have tested the specificity of the XND-1 antibody?

We included a supplementary figure demonstrating the specificity of the XND-1 antibody by Western blot. This was also previously published (Wagner et al., 2010)

(2) Figure 1D - can the authors provide an explanation why the him-5p::him-5 transgene that drives a higher expression than pie-1p::him-5 fails to suppress the Him phenotype seen in him-17? What are the HIM-5 levels like in these two strains compared to N2 and him-17 null mutants? Can this information provide explanation for the differential effect of the him-5 transgenes?

We previously reported that him-5p::him-5 drives higher expression than pie-1p::him-5 (McClendon et al, 2016).

The reason that him-5p::him-5 does not rescue, despite higher wild type expression is that HIM-17 directly regulates expression of him-5. Since HIM-17 does not regulate the pie-1 promoter, the pie-1p::him-5 construct can at least partially suppress the him-17 mutation.

We have (hopefully) explained this better in the text.

(3) Line 102- the subheading "HIM-5 is the essential factor for meiotic breaks in the Xchromosome" may not be appropriate for this section. This is what has previously been known. However, the results in Figure 1 demonstrate that a him-5 transgene can partially rescue the him-17 and -xnd-1 phenotype, but not that it is essential for meiotic DSB formation on X chromosomes.

We think some of the concern here is semantic and have changed the phraseology to say that HIM-5 is SUFFICIENT for DSBs on the X... which had not previously been shown.

Vis-à-vis the X chromosome, in all genetic backgrounds examined, the absence of HIM-5 consistently results in a complete lack of DSBs on the X. For instance, in *dsb-2* mutants—where HIM-5 is still expressed—DSBs are reduced genome-wide, but the X chromosome occasionally retains breaks. In contrast, even a weak allele of *him-17* results specifically in the loss of X chromosome breaks, underscoring a unique requirement for HIM-5 in promoting DSBs on the X. While Figure 1 shows that a *him-5* transgene can partially rescue *him-17* and *xnd-1* phenotypes, the consistent observation that X breaks are absent without HIM-5 supports its classification as sufficient for DSB formation on the X chromosome.

(4) Figure 1E - please consider enlarging the images and showing multiple examples.

Done.

I also suggest that the authors perform a more rigorous analysis to support the conclusion that *XND-1* and *HIM-17* localize away from the axis by quantifying multiple images and doing line-scan analysis.

Provided. New images are provided in both, the main and supplemental figures, and quantification is included. There is no detectable overlap of the two protein with one another or the DNA axes (see quantification of overlap in Fig. 1).

(5) Line 162 - This is the first mention of *DSB-1*, *DSB-2*, and *DSB-3* in the paper. *DSB-1* and *DSB-2* are *Rec114* homologs in *C. elegans* (Tesse et al., 2017), while *DSB-3* is a homolog of *Mei4* (Hinman et al., 2021). These proteins should be properly introduced in the introduction with appropriate citations.

Done. We appreciate the reviewer pointing out that this was the first reference to these genes.

(6) Line 169 - the rationale for this experiment is unclear. Why did the Y2H interaction between HIM-5 and *DSB-1* prompt the authors to test the rescue of *dsb-1* or *dsb-2* phenotypes by the ectopic expression of *him-5*? Do the authors have evidence that HIM-5 level is reduced in *dsb-1* or *dsb-2* mutants?

We have reorganized this section to better explain the motivation for looking at these interactions. We did see a difference in the localization in HIM-5 in the *dsb-1* mutant animals and we did have a sense that HIM-5 was critical for breaks on the X. We reasoned that it could have independent functions in promoting breaks that were not yet appreciated so wanted to do this experiment.

(7) Line 172 - "very slightly reduced". This claim requires statistical analysis.

We added statistical analysis, but we also removed this claim.

(8) Figures 2C and 2D - Can the authors provide an explanation why the *pie-1p::him-5* transgene fails to suppress the phenotypes in *dsb-1*, while the *him-5p::him-5* transgene can? Again, the rationale for these experiments is unclear. Because of this, the interpretation is also unclear.

The difference in rescue between the *pie-1p::him-5* and *him-5p::him-5* transgenes likely reflects differences in expression levels. As previously shown (McClendon et al., 2016), the *him-5p::him-5* construct results in significantly higher expression of HIM-5 protein compared to *pie-1p::him-5*. This elevated expression likely explains its ability to partially rescue the *dsb-1* phenotype. In contrast, the lower expression driven by the *pie-1* promoter is insufficient to

compensate for the absence of *dsb-1* function. We have clarified the rationale and interpretation of these experiments in the revised manuscript to better reflect this point.

(9) Lines 184-185 - the data for endogenously tagged *HIM-5::3xHA* are not shown anywhere in the paper. This must be shown.

We have added this in the supplemental figures.

(10) Figure 2D and 2E - what does the localization of *pie-1p::him-5::GFP (eaIs15)* and *him5p::him-5::GFP (eaIs4)* look like in wild-type and *dsb-1* mutants? Are the cytoplasmic aggregates caused by increased levels of *HIM-5* expression? Can the differential behavior of *him-5* transgenes provide explanation for differential rescues?

We now show both live and fixed images of *Phim-5::him-5::gfp* transgenes, as well as the localization of the endogenously HA-tagged *HIM-5* locus (Figure 2 and S3). In all cases, the protein is initially nuclear and then absent from meiotic nuclei with similar timing. The *Ppie1::him-5* transgene was very difficult to image due to low expression (even in wild type) so it not shown here. We presume it is the slightly elevated level of expression of the *Phim5::him-5::gfp* that can explain the differential rescue.

(11) Lines 221-222, where are the results shown? Please refer to Figure S3.

Done

(12) Figure S3 - these need statistical analyses.

Done

(13) Lines 230-231 - what about the *rec-1; parg-1; cep-1* triple mutant?

This is an excellent suggestion and not one we have not yet pursued. Given the lack of strong phenotypes in all combination of double mutants, we prioritized other experiments. However, we agree that examining the *rec-1; parg-1; cep-1* triple mutant would provide a valuable test of whether these factors act in the same pathway, and we appreciate the reviewer highlighting this potential future direction.

(14) Line 298 - I suggest the authors take a look at the AlphaFold prediction of DSB-1/DSB-2/DSB-3 and the comparison to human and budding yeast *Rec114/Mei4* complex in Guo et al., 2022 eLife, which could provide insights into the Y2H results.

We thank the reviewer for these comments and have indeed used these interactions and predicted homologies to zero in a region of interaction between these proteins that resembles what is seen in humans and yeast with a dimer of *REC114* like proteins wraps stabilizing a central *Mei4* helix. This is now shown in Figure 3H, I. Satisfyingly, this modeling predicts that a trimer comprised of 2 *DSB-1* proteins with *DSB-3* is more stable than a *DSB1-DSB-2-DSB-3* trimer. This might explain why *DSB-2* is not required in young adults and only becomes essential as *DSB-1* levels drop in older animals (Rosu et al., 2013)

(15) Can the authors introduce mutations within the *DSB-1* interfaces that disrupt the interaction to either *SPO-11* or *DSB-2*?

We have begun to address this question by introducing targeted mutations within *DSB-1*. As shown in Figure 3E and 3F, mutations in the C-terminal region of *DSB-1*—which includes a core of four α -helices—disrupt its interaction with *DSB-2* and *DSB-3*, but not with *SPO-11*.

These findings suggest that the C-terminus mediates interactions specifically with DSB2 and DSB-3

(16) Line 323 - The him-5 phenotypes are too weak to support the idea that it serves as the linchpin for the whole DSB complex. Do the authors have an explanation for why him-5 mutants exhibit X-chromosome-specific DSB defects?

In response to the reviewer, above, and in the text, we have included a more detailed explanation of why we think HIM-5 has a key role in coordinating meiotic break formation. Although, identified for its role on the X, the phenotypes associated with DSB formation in the mutant are really quite pleiotropic and severe.

(17) Line 436 - C. elegans lacks DSB hotspots.

Removed

Minor comments:

(1) Figure 1A - please show the raw data for the yeast two-hybrid.

We show representative yeast colonies in Figure S3.

(2) It looks like the labeling for Figure 1B and 1C are switched.

Fixed.

(3) Figure 1B - what does the red box indicate? Please explain it in the legend.

It indicates the XND-1 band. We added that information in the legend.

(4) Figure 1C - in the legend, it was noted that the results are from GFP pull-downs of HIM17::GFP. However, the method for Figure 1B and the method section noted that HIM-17 was tagged with 3xHA, and the pull-down was performed using anti-HA affinity matrix. Please reconcile this discrepancy.

That's because they were done in two different sets of experiments. For the IPs we used a HIM-17::HA strain and for the MS, a HIM-17::GFP strain.

(5) Also in Figure 1C - please call Table S2 in the main text when discussing the mass spec results. Also, it is not clear what HIM-17 and GFP indicate in the table. What makes CKU80 different from the other proteins listed under GFP? Please explain more clearly in the legend.

We have move the table to supplemental data where we have included all of the peptide counts and gene coverage. We have included in the revised method rationale for inclusion in this table which explains why CKU-80 differs.

(6) Line 527 - it is unclear what experiment was done for HIM-17. Please revise it to indicate that this is for "HIM-17 immunoprecipitation". Also please indicate the strain used for HIM17 pull-down (AV280?).

(7) Line 113- please be specific about how the HIM-17 IP was performed. Which epitope and strains are used for pull-downs?

This indeed was AV280. This has been added to the text and methods.

(8) Figure 1D- What does ND mean? In the text, it was stated that there was only a minor suppression of hatching rates. The hatching rate for *him-5p::him-5; him-17* must have been measured, and the data must be presented.

ND does mean not determined. We have removed the statement about “minor suppression”. We only tested the overall population dynamics in the *Phim-5::him-5;him17(ok424)* and the DAPI body counts. The failure to suppress the latter suggests there would be no effect on hatching rates, although we did not test this directly. Since we had done this for the *Ppie-1::him-5;him-17* strain, we provided this information to further support the claims of genetic rescue by ectopic expression.

(9) Line 151 - please specify that STED was used.

We have removed the STED images, and just show the confocal images with Lightning Processing.

(10) Figure 1E- the authors suggested that *HIM-17* and *XND-1* mainly localize to autosomes but not the X chromosome. However, there is not enough evidence that the chromosome excluded from *HIM-17* staining is indeed an X chromosome.

(11) Figure 1E (Line 154) - what are the active chromatin markers examined? Where are the data?

We have previously shown that the chromosome lacking *XND-1* staining is the X (Wagner et al., 2010). The X is heterochromatic and chromatin marks associated with active transcription, including *H3K4me3* and *HTZ-1* (a variant *H2A*), preferentially localize to autosomes, effectively anti-marking the X chromosome. As shown in the new Figure 1E, a single chromosome has very little *XND-1* and *HIM-17* associated proteins. This is the X chromosome.

(12) Line 172 - It should be a comma instead of the period after “*In dsb-1 mutants*”.

Fixed

(13) Figure S3H-K - I suggest the authors indicate the alleles of *mre-11* (null vs. *iow1*) on the graph, similarly to *him-5(e1490)* to avoid confusion.

Done

(14) Lines 294 and 600 - Guo et al. 2022 is now published in eLife. The authors must cite the published paper, not the preprint.

Fixed

(15) Line 407 - the reference Carelli et al., 2022 is missing.

Added

(16) Line 766 - please remove “is” before nuclear.

Done

Reviewer #3 (Recommendations For The Authors):

Major issues:

In my view, the most interesting mechanistic finding in the paper is the evidence that HIM-5 may not bind to chromatin in the absence of DSB-1. If validated, this would suggest that HIM-5 is likely to be directly involved in a process that promotes break formation, in contrast to factors such as HIM-17 and XND-1. It does not, however, support the idea that HIM-5 is at the top of a hierarchy of DSB factors, as it is interpreted here. More importantly, the data supporting this claim are unconvincing; only a single image of an unfixed gonad from an animal expressing HIM-5::GFP is shown. Immunofluorescence should be performed and the results must be quantified.

We have provided additional images of the HIM-5 relocalization to show that we observed this in both fixed and live worms with two different tagged strains. The exclusion from the nucleus is seen in all scenarios. Whether the protein now accumulates exclusively in the cytoplasm/ is destabilized is challenging to address with the fixed images due to the arbitrariness of defining “background” staining.

More generally, this type of analysis, looking at the interdependence of different factors for their association with chromosomes, is much more informative than the genetic interaction data presented in the paper, which does not seem to provide any mechanistic insights into the functions of the factors analyzed. The paper could potentially be greatly improved through a more extensive, systematic analysis of the interdependence of DSBpromoting factors for their localization to chromosomes.

We have at least added this for XND-1 and HIM-17 and show they are not interdependent for chromosome association. We also provide for the first time data on the localization of HIM-5 in the dsb-1 mutant. Many of the other interactions have already been shown in the literature and/or were not warranted base on the lack of genetic interaction we present here.

Minor issues:

The title is vague and inconclusive. A more concrete title summarizing the major findings would help readers to assess whether the work is of interest.

We have discussed the title extensively with all authors and all would like to keep the current title.

The authors claim that the expression of HIM-5 from a different promoter (Ppie-1::him-5) but not its endogenous promoter (Phim-5::him-5) can partially rescue the DSB defect in him-17 mutants. To support this claim, they should really quantify the germline expression of HIM-5 in wild-type, him-17, him-17; Ppie-1::him-5, and Phim-5::him-5; him-17.

We had previously reported the expression in the N2 background of both transgenes (McClendon et al., 2016)

Panel O appears to be missing from Figure S3.

Fixed

The evidence for chromosome fusions in cep-1; mre-11 mutants shown in S4D is not convincing and the claim should be removed unless stronger evidence can be obtained.

A clearer image has been added

The basis of the following statement is unclear: "Furthermore, rec-1;him-5 double mutants give an age-dependent severe loss of DSBs (like dsb-2 mutants) suggesting that the ancestral function of the protein may have a more profound effect on break formation." The manuscript does not seem to include data regarding age-dependent loss of DSBs and no other publication is cited to support this claim. The interpretation is also perplexing; I think that it may be predicated on the idea that REC-1 and HIM-5 are paralogs, but as stated above, this claim is not well supported and is likely specious.

We have added the reference. This was shown in Chung et al., 2013 – the paper that presented the cloning of the rec-1 locus.

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